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(54) **METHOD AND APPARATUS FOR STOPPING MULTICAST AND BROADCAST SERVICE IN A WIRELESS COMMUNICATION SYSTEM**

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(58) **Field of Classification Search**

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See application file for complete search history.

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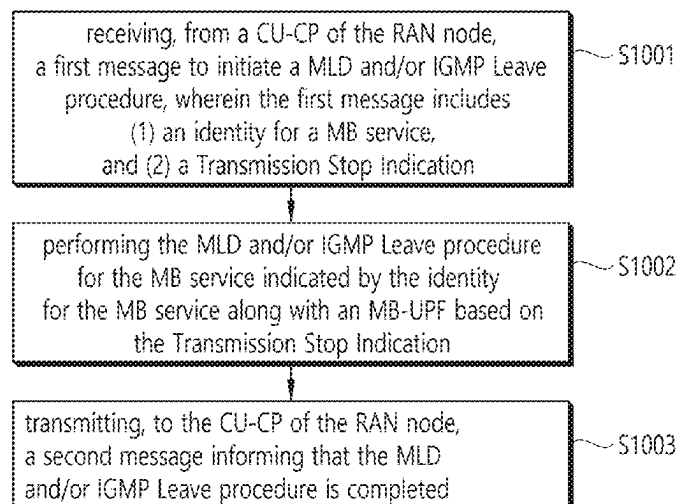
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(57) **ABSTRACT**

A method and apparatus for stopping multicast and broadcast service in a wireless communication system is provided. A CU-UP of a RAN node receives, from a CU-CP of the RAN node, a first message to initiate a MLD and/or IGMP Leave procedure. A CU-UP of a RAN node performs the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-UPF based on the Transmission Stop Indication. A CU-UP of a RAN node transmits, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

20 Claims, 12 Drawing Sheets



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H04W 76/40 (2018.01)

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FIG. 1

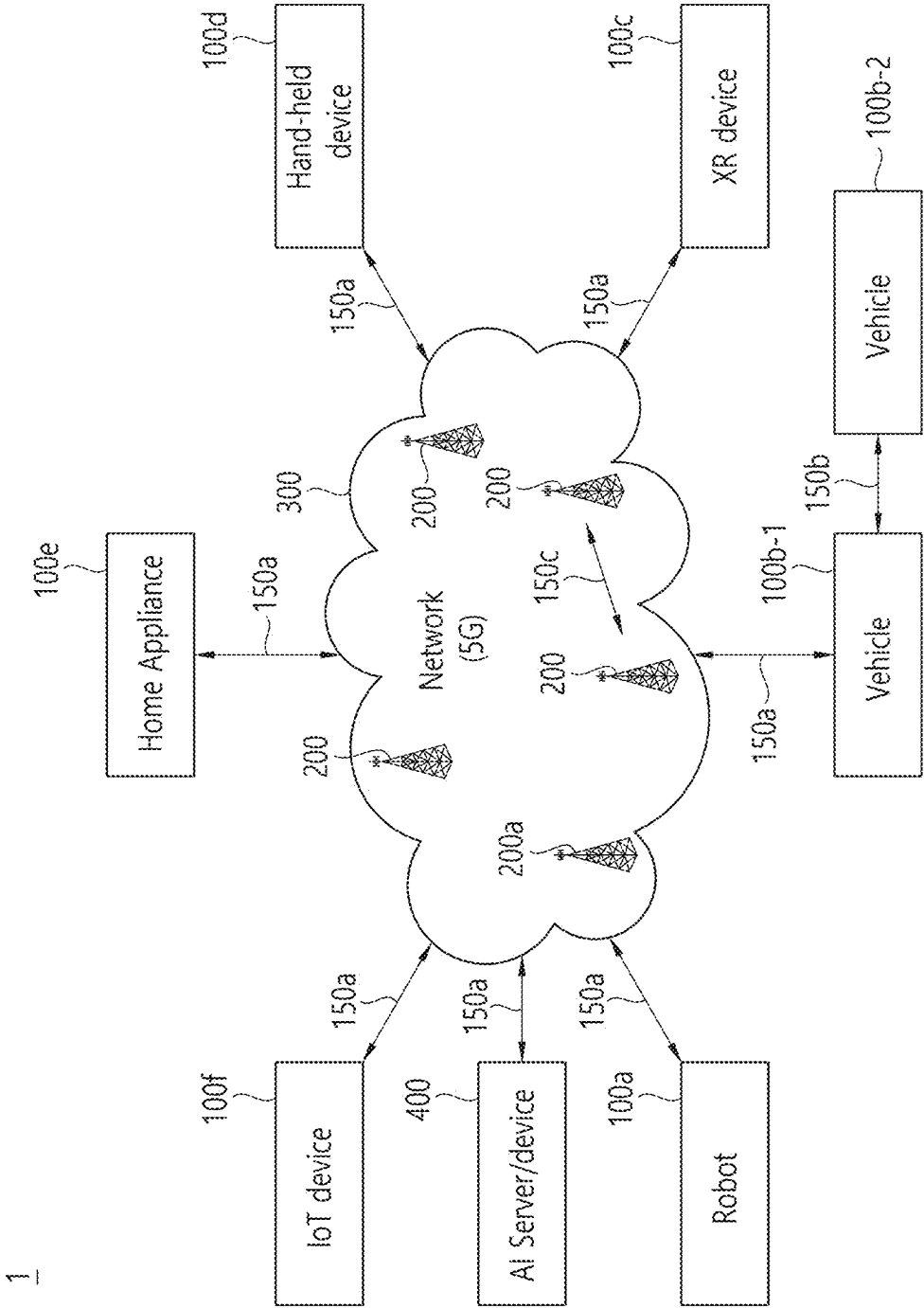


FIG. 2

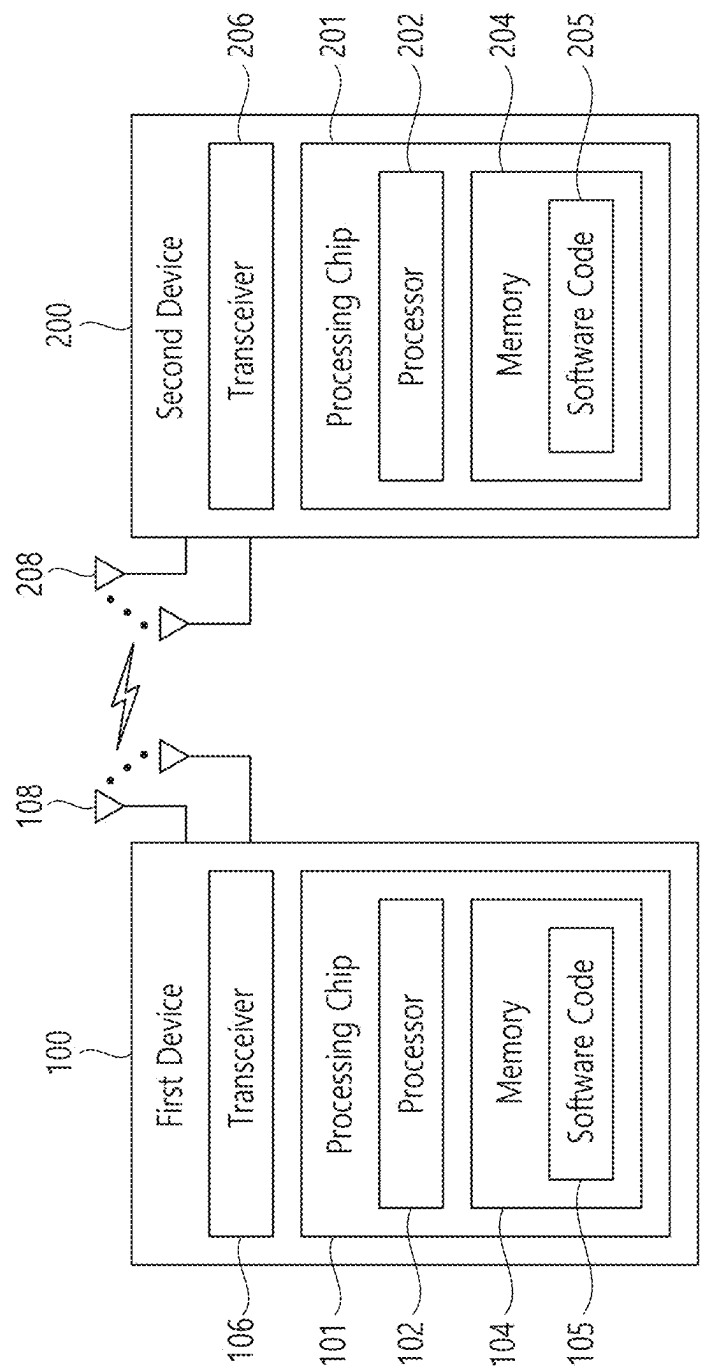


FIG. 3

Device (100,200)

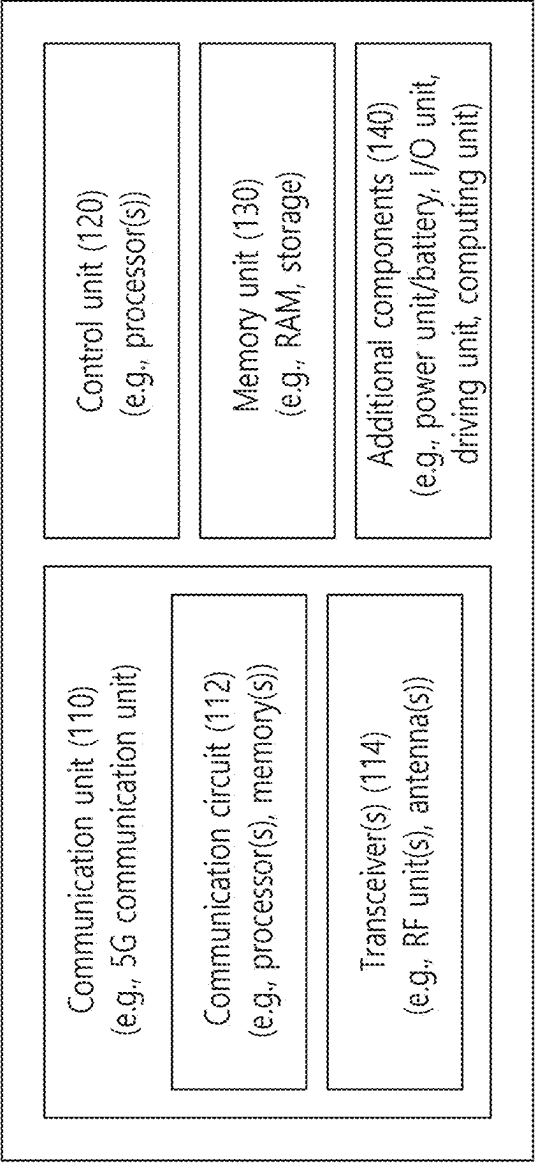


FIG. 4

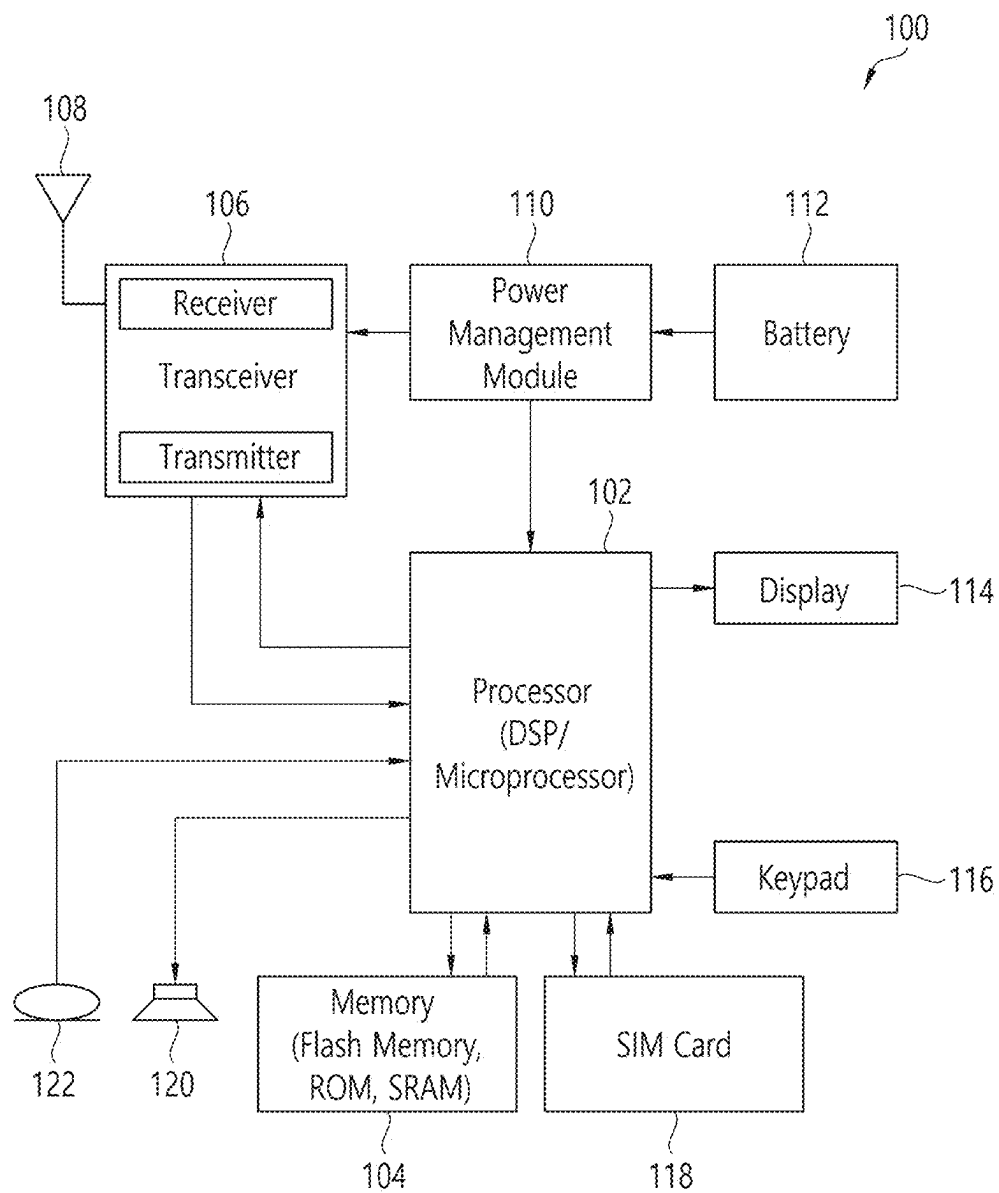


FIG. 5

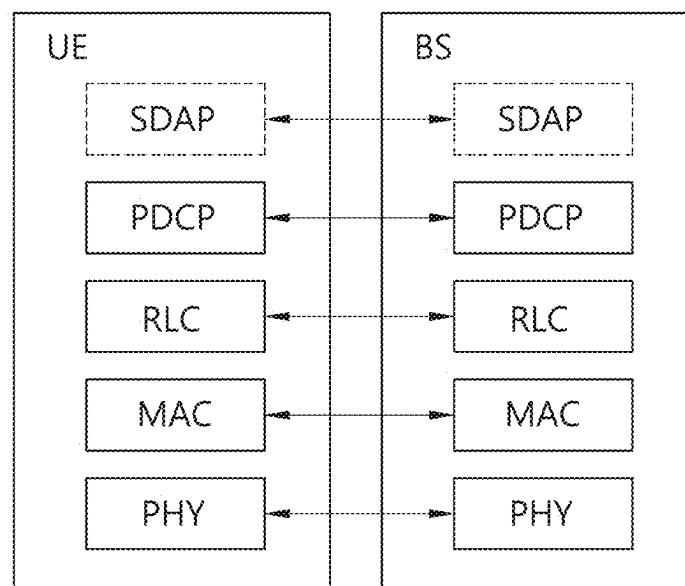


FIG. 6

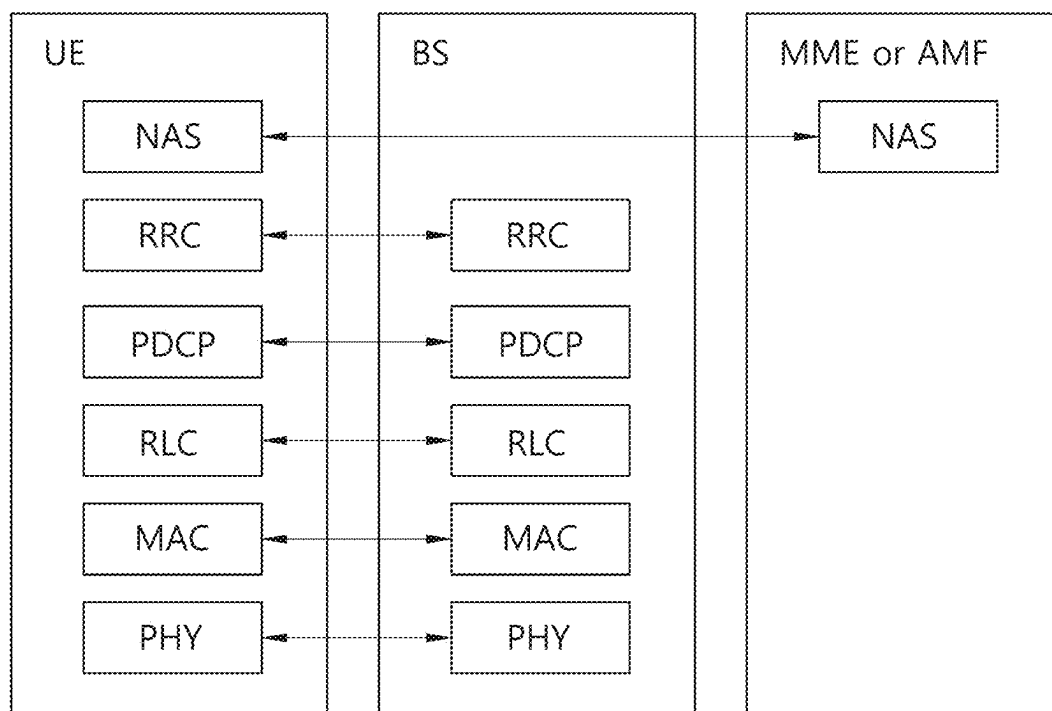


FIG. 7

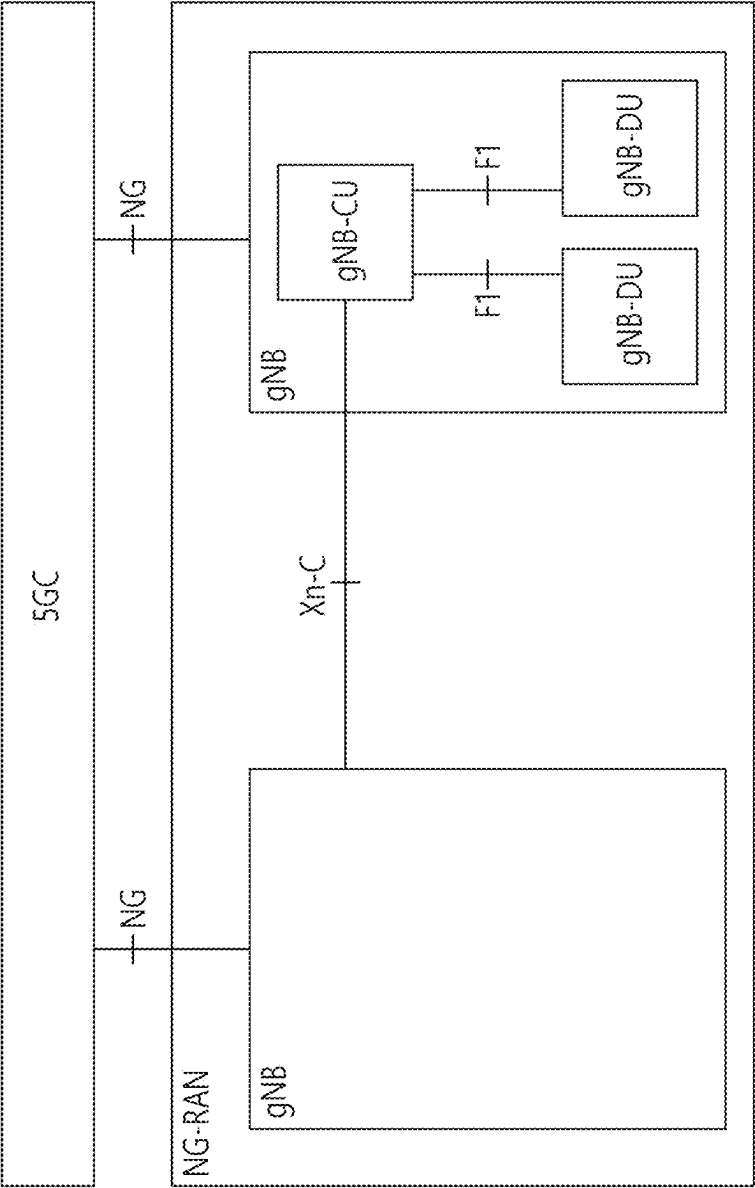


FIG. 8

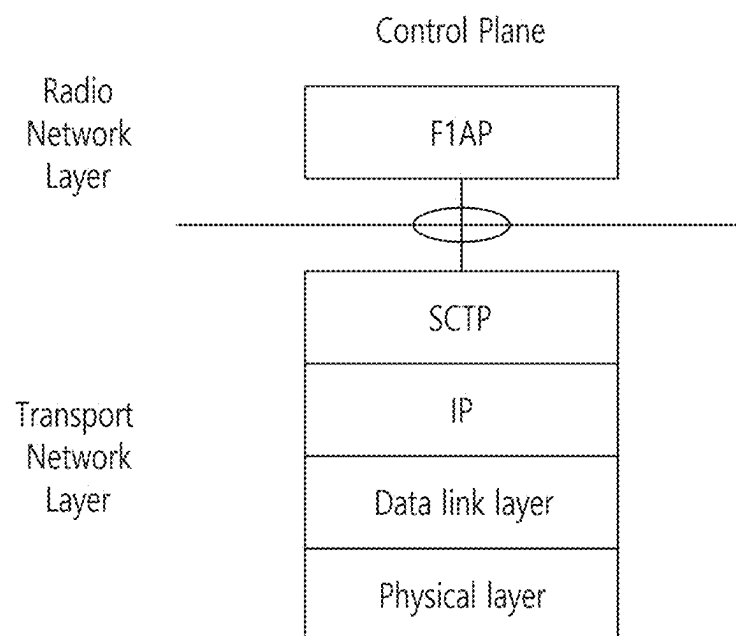


FIG. 9

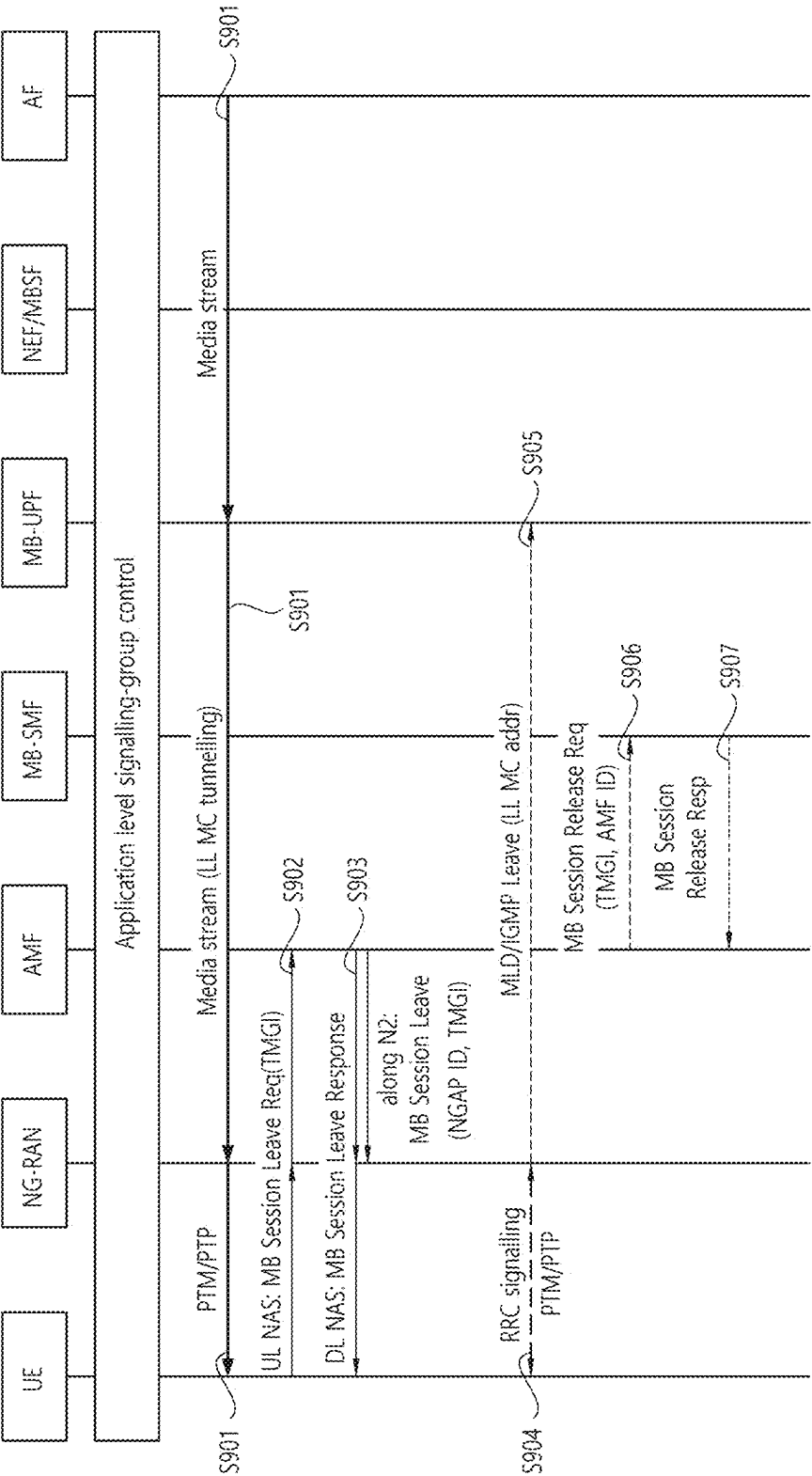


FIG. 10

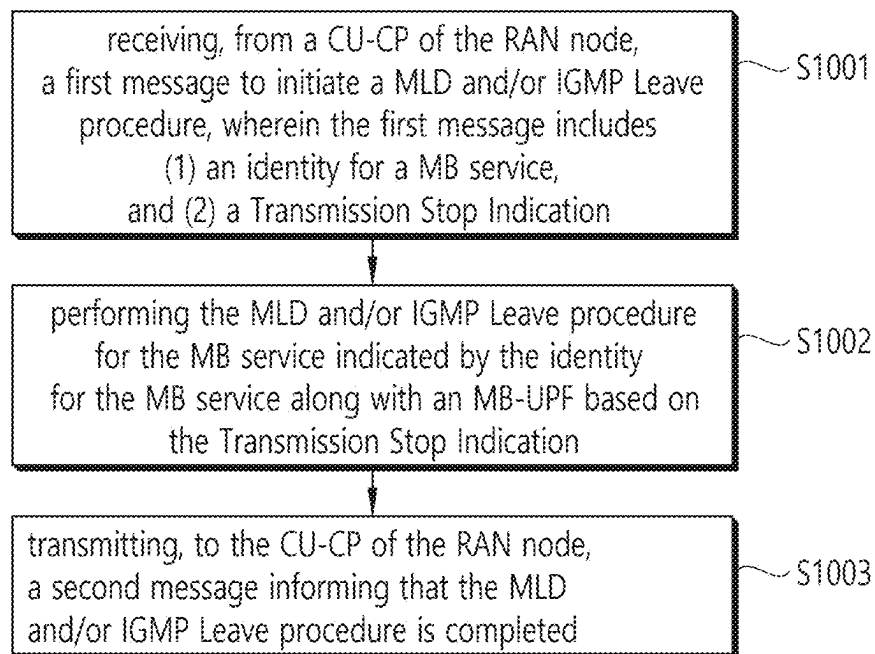


FIG. 11A

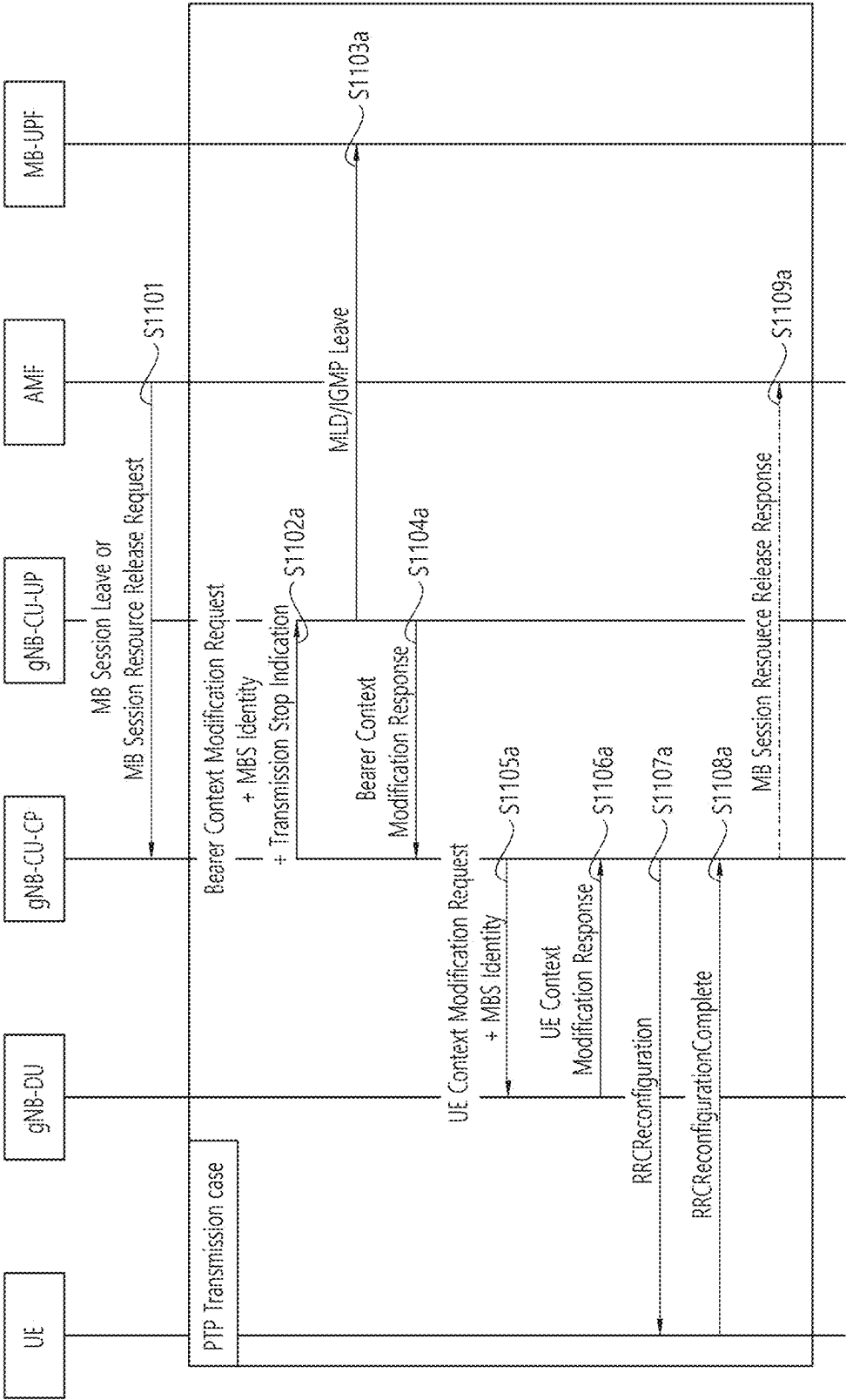
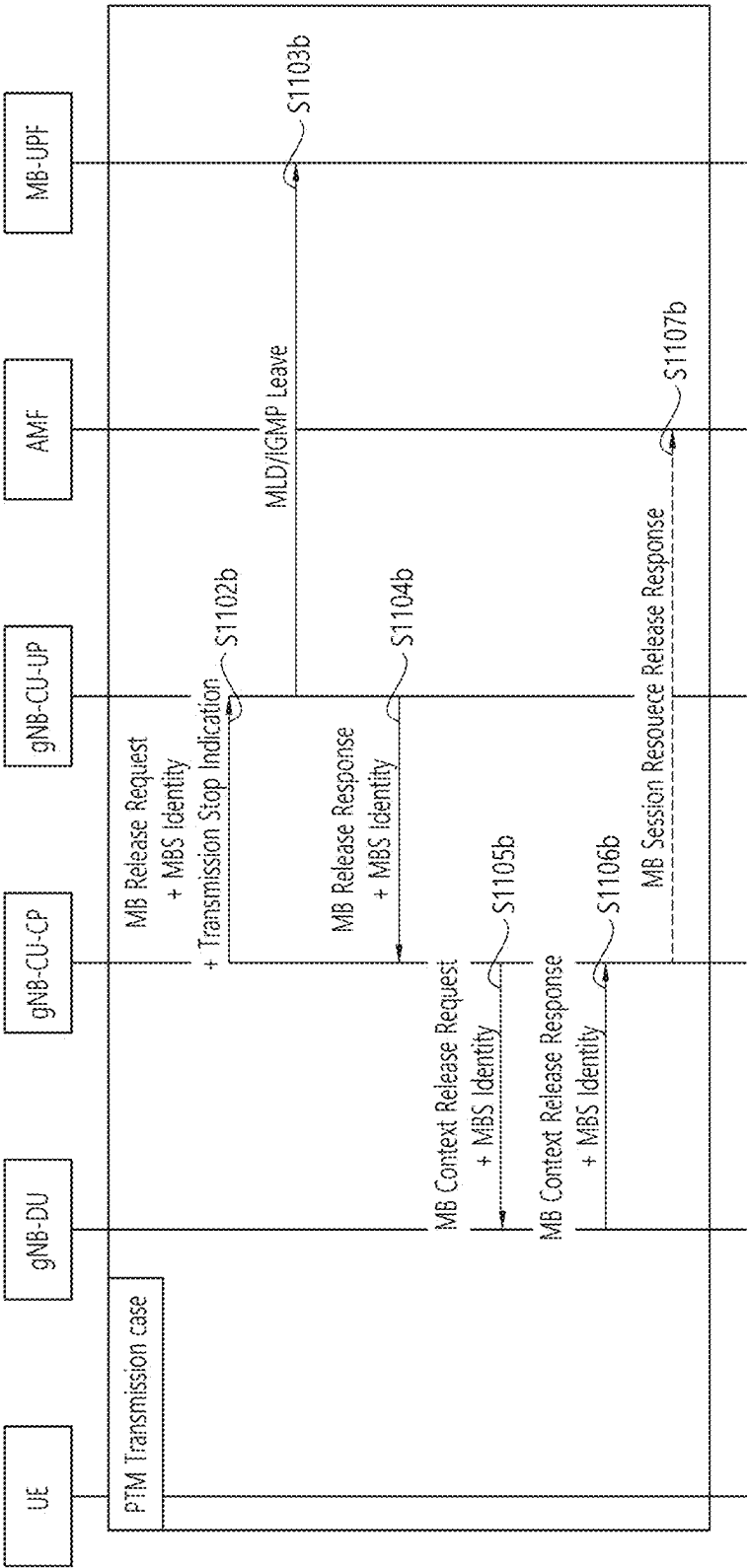


FIG. 11B



METHOD AND APPARATUS FOR STOPPING MULTICAST AND BROADCAST SERVICE IN A WIRELESS COMMUNICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/KR2021/009041, filed on Jul. 14, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0088327, filed on Jul. 16, 2020. The disclosures of the prior applications are incorporated by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to a method and apparatus for stopping multicast and broadcast service in a wireless communication system.

BACKGROUND ART

3rd generation partnership project (3GPP) long-term evolution (LTE) is a technology for enabling high-speed packet communications. Many schemes have been proposed for the LTE objective including those that aim to reduce user and provider costs, improve service quality, and expand and improve coverage and system capacity. The 3GPP LTE requires reduced cost per bit, increased service availability, flexible use of a frequency band, a simple structure, an open interface, and adequate power consumption of a terminal as an upper-level requirement.

Work has started in international telecommunication union (ITU) and 3GPP to develop requirements and specifications for new radio (NR) systems. 3GPP has to identify and develop the technology components needed for successfully standardizing the new RAT timely satisfying both the urgent market needs, and the more long-term requirements set forth by the ITU radio communication sector (ITU-R) international mobile telecommunications (IMT)-2020 process. Further, the NR should be able to use any spectrum band ranging at least up to 100 GHz that may be made available for wireless communications even in a more distant future.

The NR targets a single technical framework addressing all usage scenarios, requirements and deployment scenarios including enhanced mobile broadband (eMBB), massive machine-type-communications (mMTC), ultra-reliable and low latency communications (URLLC), etc. The NR shall be inherently forward compatible.

In NR, Multicast and Broadcast Services (MBS) may be supported.

For example, RAN basic functions may be support broadcast and/or multicast for UEs in RRC_CONNECTED state. For example, a group scheduling mechanism may be supported for UEs to receive Broadcast/Multicast service. In addition, the group scheduling mechanism may be supported to enable simultaneous operation with unicast reception. For example, dynamic change of Broadcast/Multicast service delivery between multicast (for example, Point to Multipoint (PTM)) and unicast (for example, Point to Point (PTP)) may be supported for service continuity for a given UE. For example, basic mobility may be supported for service continuity. For example, dynamic control of the Broadcast/Multicast transmission area within one Distributed Unit (DU) of a RAN node (for example, a gNodeB (gNB)) may be supported.

DISCLOSURE OF INVENTION

Technical Problem

A RAN node (for example, a gNB) may be composed of a Central Unit (CU) and a Distributed Unit (DU). For Multicast and/or Broadcast Services, one or more necessary coordination functions (for example, functions hosted by MCE) may be reside in a Central Unit (CU) of a RAN node.

A session leave procedure may be used by UEs to inform a network that the UE interest in an MB session has ceased. If a RAN node may be composed of a CU-Control Plane (CP), a CU-User Plane (UP), and a DU, signalling between the CU-CP and the CU-UP in order to request the MB-User Plane Function (UPF) to stop the media stream to above the RAN node is needed.

Therefore, studies for stopping multicast and broadcast service in a wireless communication system are needed.

Solution to Problem

In an aspect, a method performed by a Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node in a wireless communication system, is provided. A CU-UP of a RAN node receives, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure. A CU-UP of a RAN node performs the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication. A CU-UP of a RAN node transmits, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

In another aspect, an apparatus for implementing the above method is provided.

Advantageous Effects of Invention

The present disclosure may have various advantageous effects.

According to some embodiments of the present disclosure, a Radio Access Network (RAN) node (for example, a base station such as an eNB or a gNB) could efficiently stop to provide a multicast and broadcast service.

For example, a CU-UP can properly request an MB-UPF to stop multicast and/or broadcast data transmission for a MB service when a CU-CP should delete a MB session context due to session leave, stop, or delete.

For example, a RAN node could efficiently use resource for multicast and/or broadcast service.

For example, a RAN node could prevent that the UE receives multicast and/or broadcast data for the MB service in which it is not interested.

According to some embodiments of the present disclosure, a wireless communication system could provide an efficient solution for stopping a multicast and broadcast service.

Advantageous effects which can be obtained through specific embodiments of the present disclosure are not limited to the advantageous effects listed above. For example, there may be a variety of technical effects that a person having ordinary skill in the related art can understand and/or derive from the present disclosure. Accordingly, the specific effects of the present disclosure are not limited to those explicitly described herein, but may include various

effects that may be understood or derived from the technical features of the present disclosure.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows an example of a communication system to which implementations of the present disclosure is applied.

FIG. 2 shows an example of wireless devices to which implementations of the present disclosure is applied.

FIG. 3 shows an example of a wireless device to which implementations of the present disclosure is applied.

FIG. 4 shows an example of UE to which implementations of the present disclosure is applied.

FIGS. 5 and 6 show an example of protocol stacks in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

FIG. 7 shows an example of the overall architecture of an NG-RAN to which technical features of the present disclosure can be applied.

FIG. 8 shows an interface protocol structure for F1-C to which technical features of the present disclosure can be applied.

FIG. 9 shows an example of a method for a Session Leave procedure to which implementations of the present disclosure is applied.

FIG. 10 shows an example of a method performed by a Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node in a wireless communication system.

FIGS. 11A and 11B show an example of a method for stopping multicast and broadcast service in a wireless communication system.

MODE FOR THE INVENTION

The following techniques, apparatuses, and systems may be applied to a variety of wireless multiple access systems. Examples of the multiple access systems include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, a single carrier frequency division multiple access (SC-FDMA) system, and a multicarrier frequency division multiple access (MC-FDMA) system. CDMA may be embodied through radio technology such as universal terrestrial radio access (UTRA) or CDMA2000. TDMA may be embodied through radio technology such as global system for mobile communications (GSM), general packet radio service (GPRS), or enhanced data rates for GSM evolution (EDGE). OFDMA may be embodied through radio technology such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, or evolved UTRA (E-UTRA). UTRA is a part of a universal mobile telecommunications system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is a part of evolved UMTS (E-UMTS) using E-UTRA. 3GPP LTE employs OFDMA in DL and SC-FDMA in UL. LTE-advanced (LTE-A) is an evolved version of 3GPP LTE.

For convenience of description, implementations of the present disclosure are mainly described in regards to a 3GPP based wireless communication system. However, the technical features of the present disclosure are not limited thereto. For example, although the following detailed description is given based on a mobile communication system corresponding to a 3GPP based wireless communication system, aspects of the present disclosure that are not

limited to 3GPP based wireless communication system are applicable to other mobile communication systems.

For terms and technologies which are not specifically described among the terms of and technologies employed in the present disclosure, the wireless communication standard documents published before the present disclosure may be referenced.

In the present disclosure, “A or B” may mean “only A”, “only B”, or “both A and B”. In other words, “A or B” in the present disclosure may be interpreted as “A and/or B”. For example, “A, B or C” in the present disclosure may mean “only A”, “only B”, “only C”, or “any combination of A, B and C”.

In the present disclosure, slash (/) or comma (,) may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B or C”.

In the present disclosure, “at least one of A and B” may mean “only A”, “only B” or “both A and B”. In addition, the expression “at least one of A or B” or “at least one of A and/or B” in the present disclosure may be interpreted as same as “at least one of A and B”.

In addition, in the present disclosure, “at least one of A, B and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B and C”. In addition, “at least one of A, B or C” or “at least one of A, B and/or C” may mean “at least one of A, B and C”.

Also, parentheses used in the present disclosure may mean “for example”. In detail, when it is shown as “control information (PDCCH)”, “PDCCH” may be proposed as an example of “control information”. In other words, “control information” in the present disclosure is not limited to “PDCCH”, and “PDCCH” may be proposed as an example of “control information”. In addition, even when shown as “control information (i.e., PDCCH)”, “PDCCH” may be proposed as an example of “control information”.

Technical features that are separately described in one drawing in the present disclosure may be implemented separately or simultaneously.

Although not limited thereto, various descriptions, functions, procedures, suggestions, methods and/or operational flowcharts of the present disclosure disclosed herein can be applied to various fields requiring wireless communication and/or connection (e.g., 5G) between devices.

Hereinafter, the present disclosure will be described in more detail with reference to drawings. The same reference numerals in the following drawings and/or descriptions may refer to the same and/or corresponding hardware blocks, software blocks, and/or functional blocks unless otherwise indicated.

FIG. 1 shows an example of a communication system to which implementations of the present disclosure is applied.

The 5G usage scenarios shown in FIG. 1 are only exemplary, and the technical features of the present disclosure can be applied to other 5G usage scenarios which are not shown in FIG. 1.

Three main requirement categories for 5G include (1) a category of enhanced mobile broadband (eMBB), (2) a category of massive machine type communication (mMTC), and (3) a category of ultra-reliable and low latency communications (URLLC).

Partial use cases may require a plurality of categories for optimization and other use cases may focus only upon one key performance indicator (KPI). 5G supports such various use cases using a flexible and reliable method.

eMBB far surpasses basic mobile Internet access and covers abundant bidirectional work and media and enter-

tainment applications in cloud and augmented reality. Data is one of 5G core motive forces and, in a 5G era, a dedicated voice service may not be provided for the first time. In 5G, it is expected that voice will be simply processed as an application program using data connection provided by a communication system. Main causes for increased traffic volume are due to an increase in the size of content and an increase in the number of applications requiring high data transmission rate. A streaming service (of audio and video), conversational video, and mobile Internet access will be more widely used as more devices are connected to the Internet. These many application programs require connectivity of an always turned-on state in order to push real-time information and alarm for users. Cloud storage and applications are rapidly increasing in a mobile communication platform and may be applied to both work and entertainment. The cloud storage is a special use case which accelerates growth of uplink data transmission rate. 5G is also used for remote work of cloud. When a tactile interface is used, 5G demands much lower end-to-end latency to maintain user good experience. Entertainment, for example, cloud gaming and video streaming, is another core element which increases demand for mobile broadband capability. Entertainment is essential for a smartphone and a tablet in any place including high mobility environments such as a train, a vehicle, and an airplane. Other use cases are augmented reality for entertainment and information search. In this case, the augmented reality requires very low latency and instantaneous data volume.

In addition, one of the most expected 5G use cases relates a function capable of smoothly connecting embedded sensors in all fields, i.e., mMTC. It is expected that the number of potential Internet-of-things (IoT) devices will reach 204 hundred million up to the year of 2020. An industrial IoT is one of categories of performing a main role enabling a smart city, asset tracking, smart utility, agriculture, and security infrastructure through 5G.

URLLC includes a new service that will change industry through remote control of main infrastructure and an ultra-reliable/available low-latency link such as a self-driving vehicle. A level of reliability and latency is essential to control a smart grid, automatize industry, achieve robotics, and control and adjust a drone.

5G is a means of providing streaming evaluated as a few hundred megabits per second to gigabits per second and may complement fiber-to-the-home (FTTH) and cable-based broadband (or DOCSIS). Such fast speed is needed to deliver TV in resolution of 4K or more (6K, 8K, and more), as well as virtual reality and augmented reality. Virtual reality (VR) and augmented reality (AR) applications include almost immersive sports games. A specific application program may require a special network configuration. For example, for VR games, gaming companies need to incorporate a core server into an edge network server of a network operator in order to minimize latency.

Automotive is expected to be a new important motivated force in 5G together with many use cases for mobile communication for vehicles. For example, entertainment for passengers requires high simultaneous capacity and mobile broadband with high mobility. This is because future users continue to expect connection of high quality regardless of their locations and speeds. Another use case of an automotive field is an AR dashboard. The AR dashboard causes a driver to identify an object in the dark in addition to an object seen from a front window and displays a distance from the object and a movement of the object by overlapping information talking to the driver. In the future, a wireless

module enables communication between vehicles, information exchange between a vehicle and supporting infrastructure, and information exchange between a vehicle and other connected devices (e.g., devices accompanied by a pedestrian). A safety system guides alternative courses of a behavior so that a driver may drive more safely drive, thereby lowering the danger of an accident. The next stage will be a remotely controlled or self-driven vehicle. This requires very high reliability and very fast communication between different self-driven vehicles and between a vehicle and infrastructure. In the future, a self-driven vehicle will perform all driving activities and a driver will focus only upon abnormal traffic that the vehicle cannot identify. Technical requirements of a self-driven vehicle demand ultra-low latency and ultra-high reliability so that traffic safety is increased to a level that cannot be achieved by human being.

A smart city and a smart home/building mentioned as a smart society will be embedded in a high-density wireless sensor network. A distributed network of an intelligent sensor will identify conditions for costs and energy-efficient maintenance of a city or a home. Similar configurations may be performed for respective households. All of temperature sensors, window and heating controllers, burglar alarms, and home appliances are wirelessly connected. Many of these sensors are typically low in data transmission rate, power, and cost. However, real-time HD video may be demanded by a specific type of device to perform monitoring.

Consumption and distribution of energy including heat or gas is distributed at a higher level so that automated control of the distribution sensor network is demanded. The smart grid collects information and connects the sensors to each other using digital information and communication technology so as to act according to the collected information. Since this information may include behaviors of a supply company and a consumer, the smart grid may improve distribution of fuels such as electricity by a method having efficiency, reliability, economic feasibility, production sustainability, and automation. The smart grid may also be regarded as another sensor network having low latency.

Mission critical application (e.g., e-health) is one of 5G use scenarios. A health part contains many application programs capable of enjoying benefit of mobile communication. A communication system may support remote treatment that provides clinical treatment in a faraway place. Remote treatment may aid in reducing a barrier against distance and improve access to medical services that cannot be continuously available in a faraway rural area. Remote treatment is also used to perform important treatment and save lives in an emergency situation. The wireless sensor network based on mobile communication may provide remote monitoring and sensors for parameters such as heart rate and blood pressure.

Wireless and mobile communication gradually becomes important in the field of an industrial application. Wiring is high in installation and maintenance cost. Therefore, a possibility of replacing a cable with reconstructible wireless links is an attractive opportunity in many industrial fields. However, in order to achieve this replacement, it is necessary for wireless connection to be established with latency, reliability, and capacity similar to those of the cable and management of wireless connection needs to be simplified. Low latency and a very low error probability are new requirements when connection to 5G is needed.

Logistics and freight tracking are important use cases for mobile communication that enables inventory and package tracking anywhere using a location-based information sys-

tem. The use cases of logistics and freight typically demand low data rate but require location information with a wide range and reliability.

Referring to FIG. 1, the communication system 1 includes wireless devices 100a to 100f, base stations (BSs) 200, and a network 300. Although FIG. 1 illustrates a 5G network as an example of the network of the communication system 1, the implementations of the present disclosure are not limited to the 5G system, and can be applied to the future communication system beyond the 5G system.

The BSs 200 and the network 300 may be implemented as wireless devices and a specific wireless device may operate as a BS/network node with respect to other wireless devices.

The wireless devices 100a to 100f represent devices performing communication using radio access technology (RAT) (e.g., 5G new RAT (NR)) or LTE) and may be referred to as communication/radio/5G devices. The wireless devices 100a to 100f may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an extended reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an IoT device 100f, and an artificial intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. The vehicles may include an unmanned aerial vehicle (UAV) (e.g., a drone). The XR device may include an AR/VR/Mixed Reality (MR) device and may be implemented in the form of a head-mounted device (HMD), a head-up display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smart-pad, a wearable device (e.g., a smartwatch or a smart-glasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter.

In the present disclosure, the wireless devices 100a to 100f may be called user equipments (UEs). A UE may include, for example, a cellular phone, a smartphone, a laptop computer, a digital broadcast terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation system, a slate personal computer (PC), a tablet PC, an ultrabook, a vehicle, a vehicle having an autonomous traveling function, a connected car, an UAV, an AI module, a robot, an AR device, a VR device, an MR device, a hologram device, a public safety device, an MTC device, an IoT device, a medical device, a FinTech device (or a financial device), a security device, a weather/environment device, a device related to a 5G service, or a device related to a fourth industrial revolution field.

The UAV may be, for example, an aircraft aviated by a wireless control signal without a human being onboard.

The VR device may include, for example, a device for implementing an object or a background of the virtual world. The AR device may include, for example, a device implemented by connecting an object or a background of the virtual world to an object or a background of the real world. The MR device may include, for example, a device implemented by merging an object or a background of the virtual world into an object or a background of the real world. The hologram device may include, for example, a device for implementing a stereoscopic image of 360 degrees by recording and re-producing stereoscopic information, using an interference phenomenon of light generated when two laser lights called holography meet.

The public safety device may include, for example, an image relay device or an image device that is wearable on the body of a user.

The MTC device and the IoT device may be, for example, devices that do not require direct human intervention or manipulation. For example, the MTC device and the IoT device may include smartmeters, vending machines, thermometers, smartbulbs, door locks, or various sensors.

The medical device may be, for example, a device used for the purpose of diagnosing, treating, relieving, curing, or preventing disease. For example, the medical device may be a device used for the purpose of diagnosing, treating, relieving, or correcting injury or impairment. For example, the medical device may be a device used for the purpose of inspecting, replacing, or modifying a structure or a function. For example, the medical device may be a device used for the purpose of adjusting pregnancy. For example, the medical device may include a device for treatment, a device for operation, a device for (in vitro) diagnosis, a hearing aid, or a device for procedure.

The security device may be, for example, a device installed to prevent a danger that may arise and to maintain safety. For example, the security device may be a camera, a closed-circuit TV (CCTV), a recorder, or a black box.

The FinTech device may be, for example, a device capable of providing a financial service such as mobile payment. For example, the FinTech device may include a payment device or a point of sales (POS) system.

The weather/environment device may include, for example, a device for monitoring or predicting a weather/environment.

The wireless devices 100a to 100f may be connected to the network 300 via the BSs 200. An AI technology may be applied to the wireless devices 100a to 100f and the wireless devices 100a to 100f may be connected to the AI server 400 via the network 300. The network 300 may be configured using a 3G network, a 4G (e.g., LTE) network, a 5G (e.g., NR) network, and a beyond-5G network. Although the wireless devices 100a to 100f may communicate with each other through the BSs 200/network 300, the wireless devices 100a to 100f may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs 200/network 300. For example, the vehicles 100b-1 and 100b-2 may perform direct communication (e.g., vehicle-to-vehicle (V2V)/vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices 100a to 100f.

Wireless communication/connections 150a, 150b and 150c may be established between the wireless devices 100a to 100f and/or between wireless device 100a to 100f and BS 200 and/or between BSs 200. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication 150a, sidelink communication (or device-to-device (D2D) communication) 150b, inter-base station communication 150c (e.g., relay, integrated access and backhaul (IAB)), etc. The wireless devices 100a to 100f and the BSs 200/the wireless devices 100a to 100f may transmit/receive radio signals to/from each other through the wireless communication/connections 150a, 150b and 150c. For example, the wireless communication/connections 150a, 150b and 150c may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/de-mapping), and

resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

AI refers to the field of studying artificial intelligence or the methodology that can create it, and machine learning refers to the field of defining various problems addressed in the field of AI and the field of methodology to solve them. Machine learning is also defined as an algorithm that increases the performance of a task through steady experience on a task.

Robot means a machine that automatically processes or operates a given task by its own ability. In particular, robots with the ability to recognize the environment and make self-determination to perform actions can be called intelligent robots. Robots can be classified as industrial, medical, home, military, etc., depending on the purpose or area of use. The robot can perform a variety of physical operations, such as moving the robot joints with actuators or motors. The movable robot also includes wheels, brakes, propellers, etc., on the drive, allowing it to drive on the ground or fly in the air.

Autonomous driving means a technology that drives on its own, and autonomous vehicles mean vehicles that drive without user's control or with minimal user's control. For example, autonomous driving may include maintaining lanes in motion, automatically adjusting speed such as adaptive cruise control, automatic driving along a set route, and automatically setting a route when a destination is set. The vehicle covers vehicles equipped with internal combustion engines, hybrid vehicles equipped with internal combustion engines and electric motors, and electric vehicles equipped with electric motors, and may include trains, motorcycles, etc., as well as cars. Autonomous vehicles can be seen as robots with autonomous driving functions.

Extended reality is collectively referred to as VR, AR, and MR. VR technology provides objects and backgrounds of real world only through computer graphic (CG) images. AR technology provides a virtual CG image on top of a real object image. MR technology is a CG technology that combines and combines virtual objects into the real world. MR technology is similar to AR technology in that they show real and virtual objects together. However, there is a difference in that in AR technology, virtual objects are used as complementary forms to real objects, while in MR technology, virtual objects and real objects are used as equal personalities.

NR supports multiples numerologies (and/or multiple subcarrier spacings (SCS)) to support various 5G services. For example, if SCS is 15 kHz, wide area can be supported in traditional cellular bands, and if SCS is 30 kHz/60 kHz, dense-urban, lower latency, and wider carrier bandwidth can be supported. If SCS is 60 kHz or higher, bandwidths greater than 24.25 GHz can be supported to overcome phase noise.

The NR frequency band may be defined as two types of frequency range, i.e., FR1 and FR2. The numerical value of the frequency range may be changed. For example, the frequency ranges of the two types (FR1 and FR2) may be as shown in Table 1 below. For ease of explanation, in the frequency ranges used in the NR system, FR1 may mean "sub 6 GHz range", FR2 may mean "above 6 GHz range," and may be referred to as millimeter wave (mmW).

TABLE 1

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing
FR1	450 MHz-6000 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

As mentioned above, the numerical value of the frequency range of the NR system may be changed. For example, FR1 may include a frequency band of 410 MHz to 7125 MHz as shown in Table 2 below. That is, FR1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, etc.) or more. For example, a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, etc.) or more included in FR1 may include an unlicensed band. Unlicensed bands may be used for a variety of purposes, for example for communication for vehicles (e.g., autonomous driving).

TABLE 2

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

Here, the radio communication technologies implemented in the wireless devices in the present disclosure may include narrowband internet-of-things (NB-IoT) technology for low-power communication as well as LTE, NR and 6G. For example, NB-IoT technology may be an example of low power wide area network (LPWAN) technology, may be implemented in specifications such as LTE Cat NB1 and/or LTE Cat NB2, and may not be limited to the above-mentioned names. Additionally and/or alternatively, the radio communication technologies implemented in the wireless devices in the present disclosure may communicate based on LTE-M technology. For example, LTE-M technology may be an example of LPWAN technology and be called by various names such as enhanced machine type communication (eMTC). For example, LTE-M technology may be implemented in at least one of the various specifications, such as 1) LTE Cat 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-bandwidth limited (non-BL), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and may not be limited to the above-mentioned names. Additionally and/or alternatively, the radio communication technologies implemented in the wireless devices in the present disclosure may include at least one of ZigBee, Bluetooth, and/or LPWAN which take into account low-power communication, and may not be limited to the above-mentioned names. For example, ZigBee technology may generate personal area networks (PANs) associated with small/low-power digital communication based on various specifications such as IEEE 802.15.4 and may be called various names.

FIG. 2 shows an example of wireless devices to which implementations of the present disclosure is applied.

Referring to FIG. 2, a first wireless device 100 and a second wireless device 200 may transmit/receive radio signals to/from an external device through a variety of RATs (e.g., LTE and NR).

In FIG. 2, {the first wireless device 100 and the second wireless device 200} may correspond to at least one of {the wireless device 100a to 100f and the BS 200}, {the wireless device 100a to 100f and the wireless device 100a to 100f} and/or {the BS 200 and the BS 200} of FIG. 1.

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The first wireless device **100** may include at least one transceiver, such as a transceiver **106**, at least one processing chip, such as a processing chip **101**, and/or one or more antennas **108**.

The processing chip **101** may include at least one processor, such as a processor **102**, and at least one memory, such as a memory **104**. It is exemplarily shown in FIG. 2 that the memory **104** is included in the processing chip **101**. Additional and/or alternatively, the memory **104** may be placed outside of the processing chip **101**.

The processor **102** may control the memory **104** and/or the transceiver **106** and may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. For example, the processor **102** may process information within the memory **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver **106**. The processor **102** may receive radio signals including second information/signals through the transceiver **106** and then store information obtained by processing the second information/signals in the memory **104**.

The memory **104** may be operably connectable to the processor **102**. The memory **104** may store various types of information and/or instructions. The memory **104** may store a software code **105** which implements instructions that, when executed by the processor **102**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **105** may implement instructions that, when executed by the processor **102**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **105** may control the processor **102** to perform one or more protocols. For example, the software code **105** may control the processor **102** to perform one or more layers of the radio interface protocol.

Herein, the processor **102** and the memory **104** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver **106** may be connected to the processor **102** and transmit and/or receive radio signals through one or more antennas **108**. Each of the transceiver **106** may include a transmitter and/or a receiver. The transceiver **106** may be interchangeably used with radio frequency (RF) unit(s). In the present disclosure, the first wireless device **100** may represent a communication modem/circuit/chip.

The second wireless device **200** may include at least one transceiver, such as a transceiver **206**, at least one processing chip, such as a processing chip **201**, and/or one or more antennas **208**.

The processing chip **201** may include at least one processor, such as a processor **202**, and at least one memory, such as a memory **204**. It is exemplarily shown in FIG. 2 that the memory **204** is included in the processing chip **201**. Additional and/or alternatively, the memory **204** may be placed outside of the processing chip **201**.

The processor **202** may control the memory **204** and/or the transceiver **206** and may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. For example, the processor **202** may process information within the memory **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver **206**. The processor **202** may receive radio signals including fourth

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information/signals through the transceiver **106** and then store information obtained by processing the fourth information/signals in the memory **204**.

The memory **204** may be operably connectable to the processor **202**. The memory **204** may store various types of information and/or instructions. The memory **204** may store a software code **205** which implements instructions that, when executed by the processor **202**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **205** may implement instructions that, when executed by the processor **202**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **205** may control the processor **202** to perform one or more protocols. For example, the software code **205** may control the processor **202** to perform one or more layers of the radio interface protocol.

Herein, the processor **202** and the memory **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver **206** may be connected to the processor **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver **206** may include a transmitter and/or a receiver. The transceiver **206** may be interchangeably used with RF unit. In the present disclosure, the second wireless device **200** may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may implement one or more layers (e.g., functional layers such as physical (PHY) layer, media access control (MAC) layer, radio link control (RLC) layer, packet data convergence protocol (PDCP) layer, radio resource control (RRC) layer, and service data adaptation protocol (SDAP) layer). The one or more processors **102** and **202** may generate one or more protocol data units (PDUs) and/or one or more service data unit (SDUs) according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure.

The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more application specific integrated circuits (ASICs), one or more digital

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signal processors (DSPs), one or more digital signal processing devices (DSPDs), one or more programmable logic devices (PLDs), or one or more field programmable gate arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by read-only memories (ROMs), random access memories (RAMs), electrically erasable programmable read-only memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices.

The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, through the one or more antennas **108** and **208**. In the present disclosure, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports).

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The one or more transceivers **106** and **206** may convert received radio signals/channels, etc., from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc., using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc., processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters. For example, the transceivers **106** and **206** can up-convert OFDM baseband signals to a carrier frequency by their (analog) oscillators and/or filters under the control of the processors **102** and **202** and transmit the up-converted OFDM signals at the carrier frequency. The transceivers **106** and **206** may receive OFDM signals at a carrier frequency and down-convert the OFDM signals into OFDM baseband signals by their (analog) oscillators and/or filters under the control of the transceivers **102** and **202**.

In the implementations of the present disclosure, a UE may operate as a transmitting device in uplink (UL) and as a receiving device in downlink (DL). In the implementations of the present disclosure, a BS may operate as a receiving device in UL and as a transmitting device in DL. Hereinafter, for convenience of description, it is mainly assumed that the first wireless device **100** acts as the UE, and the second wireless device **200** acts as the BS. For example, the processor(s) **102** connected to, mounted on or launched in the first wireless device **100** may be configured to perform the UE behavior according to an implementation of the present disclosure or control the transceiver(s) **106** to perform the UE behavior according to an implementation of the present disclosure. The processor(s) **202** connected to, mounted on or launched in the second wireless device **200** may be configured to perform the BS behavior according to an implementation of the present disclosure or control the transceiver(s) **206** to perform the BS behavior according to an implementation of the present disclosure.

In the present disclosure, a BS is also referred to as a node B (NB), an eNode B (eNB), or a gNB.

FIG. 3 shows an example of a wireless device to which implementations of the present disclosure is applied.

The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 1).

Referring to FIG. 3, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 2 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit **110** may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** of FIG. 2 and/or the one or more memories **104** and **204** of FIG. 2. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** of FIG. 2 and/or the one or more antennas **108** and **208** of FIG. 2. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of each of the wireless devices **100** and **200**. For example, the control unit **120** may control an electric/mechanical operation of each of the wireless devices **100** and **200** based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the

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communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of the wireless devices **100** and **200**. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit (e.g., audio I/O port, video I/O port), a driving unit, and a computing unit. The wireless devices **100** and **200** may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 1), the vehicles (**100b-1** and **100b-2** of FIG. 1), the XR device (**100c** of FIG. 1), the hand-held device (**100d** of FIG. 1), the home appliance (**100e** of FIG. 1), the IoT device (**100f** of FIG. 1), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a FinTech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. 1), the BSs (**200** of FIG. 1), a network node, etc. The wireless devices **100** and **200** may be used in a mobile or fixed place according to a use-example/service.

In FIG. 3, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor (AP), an electronic control unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a RAM, a DRAM, a ROM, a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

FIG. 4 shows an example of UE to which implementations of the present disclosure is applied.

Referring to FIG. 4, a UE **100** may correspond to the first wireless device **100** of FIG. 2 and/or the wireless device **100** or **200** of FIG. 3.

A UE **100** includes a processor **102**, a memory **104**, a transceiver **106**, one or more antennas **108**, a power management module **110**, a battery **112**, a display **114**, a keypad **116**, a subscriber identification module (SIM) card **118**, a speaker **120**, and a microphone **122**.

The processor **102** may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The processor **102** may be configured to control one or more other components of the UE **100** to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. Layers of the radio interface protocol may be implemented in the processor **102**. The processor **102** may include ASIC, other chipset, logic circuit and/or data processing device. The processor **102** may be an application processor. The processor **102** may include at least one of a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), a modem (modu-

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lator and demodulator). An example of the processor **102** may be found in SNAPDRAGON™ series of processors made by Qualcomm®, EXYNOS™ series of processors made by Samsung®, A series of processors made by Apple®, HELIO™ series of processors made by MediaTek®, ATOM™ series of processors made by Intel® or a corresponding next generation processor.

The memory **104** is operatively coupled with the processor **102** and stores a variety of information to operate the processor **102**. The memory **104** may include ROM, RAM, flash memory, memory card, storage medium and/or other storage device. When the embodiments are implemented in software, the techniques described herein can be implemented with modules (e.g., procedures, functions, etc.) that perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The modules can be stored in the memory **104** and executed by the processor **102**. The memory **104** can be implemented within the processor **102** or external to the processor **102** in which case those can be communicatively coupled to the processor **102** via various means as is known in the art.

The transceiver **106** is operatively coupled with the processor **102**, and transmits and/or receives a radio signal. The transceiver **106** includes a transmitter and a receiver. The transceiver **106** may include baseband circuitry to process radio frequency signals. The transceiver **106** controls the one or more antennas **108** to transmit and/or receive a radio signal.

The power management module **110** manages power for the processor **102** and/or the transceiver **106**. The battery **112** supplies power to the power management module **110**.

The display **114** outputs results processed by the processor **102**. The keypad **116** receives inputs to be used by the processor **102**. The keypad **116** may be shown on the display **114**.

The SIM card **118** is an integrated circuit that is intended to securely store the international mobile subscriber identity (IMSI) number and its related key, which are used to identify and authenticate subscribers on mobile telephony devices (such as mobile phones and computers). It is also possible to store contact information on many SIM cards.

The speaker **120** outputs sound-related results processed by the processor **102**. The microphone **122** receives sound-related inputs to be used by the processor **102**.

FIGS. 5 and 6 show an example of protocol stacks in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

In particular, FIG. 5 illustrates an example of a radio interface user plane protocol stack between a UE and a BS and FIG. 6 illustrates an example of a radio interface control plane protocol stack between a UE and a BS. The control plane refers to a path through which control messages used to manage call by a UE and a network are transported. The user plane refers to a path through which data generated in an application layer, for example, voice data or Internet packet data are transported. Referring to FIG. 5, the user plane protocol stack may be divided into Layer 1 (i.e., a PHY layer) and Layer 2. Referring to FIG. 6, the control plane protocol stack may be divided into Layer 1 (i.e., a PHY layer), Layer 2, Layer 3 (e.g., an RRC layer), and a non-access stratum (NAS) layer. Layer 1, Layer 2 and Layer 3 are referred to as an access stratum (AS).

In the 3GPP LTE system, the Layer 2 is split into the following sublayers: MAC, RLC, and PDCP. In the 3GPP NR system, the Layer 2 is split into the following sublayers: MAC, RLC, PDCP and SDAP. The PHY layer offers to the

MAC sublayer transport channels, the MAC sublayer offers to the RLC sublayer logical channels, the RLC sublayer offers to the PDCP sublayer RLC channels, the PDCP sublayer offers to the SDAP sublayer radio bearers. The SDAP sublayer offers to 5G core network quality of service (QoS) flows.

In the 3GPP NR system, the main services and functions of the MAC sublayer include: mapping between logical channels and transport channels; multiplexing/de-multiplexing of MAC SDUs belonging to one or different logical channels into/from transport blocks (TB) delivered to/from the physical layer on transport channels; scheduling information reporting; error correction through hybrid automatic repeat request (HARQ) (one HARQ entity per cell in case of carrier aggregation (CA)); priority handling between UEs by means of dynamic scheduling; priority handling between logical channels of one UE by means of logical channel prioritization; padding. A single MAC entity may support multiple numerologies, transmission timings and cells. Mapping restrictions in logical channel prioritization control which numerology(ies), cell(s), and transmission timing(s) a logical channel can use.

Different kinds of data transfer services are offered by MAC. To accommodate different kinds of data transfer services, multiple types of logical channels are defined, i.e., each supporting transfer of a particular type of information. Each logical channel type is defined by what type of information is transferred. Logical channels are classified into two groups: control channels and traffic channels. Control channels are used for the transfer of control plane information only, and traffic channels are used for the transfer of user plane information only. Broadcast control channel (BCCH) is a downlink logical channel for broadcasting system control information, paging control channel (PCCH) is a downlink logical channel that transfers paging information, system information change notifications and indications of ongoing public warning service (PWS) broadcasts, common control channel (CCCH) is a logical channel for transmitting control information between UEs and network and used for UEs having no RRC connection with the network, and dedicated control channel (DCCH) is a point-to-point bi-directional logical channel that transmits dedicated control information between a UE and the network and used by UEs having an RRC connection. Dedicated traffic channel (DTCH) is a point-to-point logical channel, dedicated to one UE, for the transfer of user information. A DTCH can exist in both uplink and downlink. In downlink, the following connections between logical channels and transport channels exist: BCCH can be mapped to broadcast channel (BCH); BCCH can be mapped to downlink shared channel (DL-SCH); PCCH can be mapped to paging channel (PCH); CCCH can be mapped to DL-SCH; DCCH can be mapped to DL-SCH; and DTCH can be mapped to DL-SCH. In uplink, the following connections between logical channels and transport channels exist: CCCH can be mapped to uplink shared channel (UL-SCH); DCCH can be mapped to UL-SCH; and DTCH can be mapped to UL-SCH.

The RLC sublayer supports three transmission modes: transparent mode (TM), unacknowledged mode (UM), and acknowledged mode (AM). The RLC configuration is per logical channel with no dependency on numerologies and/or transmission durations. In the 3GPP NR system, the main services and functions of the RLC sublayer depend on the transmission mode and include: transfer of upper layer PDUs; sequence numbering independent of the one in PDCP (UM and AM); error correction through ARQ (AM only); segmentation (AM and UM) and re-segmentation (AM only)

of RLC SDUs; re-assembly of SDU (AM and UM); duplicate detection (AM only); RLC SDU discard (AM and UM); RLC re-establishment; protocol error detection (AM only).

In the 3GPP NR system, the main services and functions of the PDCP sublayer for the user plane include: sequence numbering; header compression and decompression using robust header compression (ROHC); transfer of user data; reordering and duplicate detection; in-order delivery; PDCP PDU routing (in case of split bearers); re-transmission of PDCP SDUs; ciphering, deciphering and integrity protection; PDCP SDU discard; PDCP re-establishment and data recovery for RLC AM; PDCP status reporting for RLC AM; duplication of PDCP PDUs and duplicate discard indication to lower layers. The main services and functions of the PDCP sublayer for the control plane include: sequence numbering; ciphering, deciphering and integrity protection; transfer of control plane data; reordering and duplicate detection; in-order delivery; duplication of PDCP PDUs and duplicate discard indication to lower layers.

In the 3GPP NR system, the main services and functions of SDAP include: mapping between a QoS flow and a data radio bearer; marking QoS flow ID (QFI) in both DL and UL packets. A single protocol entity of SDAP is configured for each individual PDU session.

In the 3GPP NR system, the main services and functions of the RRC sublayer include: broadcast of system information related to AS and NAS; paging initiated by 5GC or NG-RAN; establishment, maintenance and release of an RRC connection between the UE and NG-RAN; security functions including key management; establishment, configuration, maintenance and release of signaling radio bearers (SRBs) and data radio bearers (DRBs); mobility functions (including: handover and context transfer, UE cell selection and reselection and control of cell selection and reselection, inter-RAT mobility); QoS management functions; UE measurement reporting and control of the reporting; detection of and recovery from radio link failure; NAS message transfer to/from NAS from/to UE.

FIG. 7 shows an example of the overall architecture of an NG-RAN to which technical features of the present disclosure can be applied.

Referring to FIG. 7, a gNB may include a gNB-CU (hereinafter, gNB-CU may be simply referred to as CU) and at least one gNB-DU (hereinafter, gNB-DU may be simply referred to as DU).

The gNB-CU is a logical node hosting RRC, SDAP and PDCP protocols of the gNB or an RRC and PDCP protocols of the en-gNB. The gNB-CU controls the operation of the at least one gNB-DU.

The gNB-DU is a logical node hosting RLC, MAC, and physical layers of the gNB or the en-gNB. The operation of the gNB-DU is partly controlled by the gNB-CU. One gNB-DU supports one or multiple cells. One cell is supported by only one gNB-DU.

The gNB-CU and gNB-DU are connected via an F1 interface. The gNB-CU terminates the F1 interface connected to the gNB-DU. The gNB-DU terminates the F1 interface connected to the gNB-CU. One gNB-DU is connected to only one gNB-CU. However, the gNB-DU may be connected to multiple gNB-CUs by appropriate implementation. The F1 interface is a logical interface. For NG-RAN, the NG and Xn-C interfaces for a gNB consisting of a gNB-CU and gNB-DUs, terminate in the gNB-CU. For E-UTRAN-NR dual connectivity (EN-DC), the S1-U and X2-C interfaces for a gNB consisting of a gNB-CU and

gNB-DUs, terminate in the gNB-CU. The gNB-CU and connected gNB-DUs are only visible to other gNBs and the 5GC as a gNB.

Functions of the F1 interface includes F1 control (F1-C) functions as follows.

(1) F1 Interface Management Function

The error indication function is used by the gNB-DU or gNB-CU to indicate to the gNB-CU or gNB-DU that an error has occurred.

The reset function is used to initialize the peer entity after node setup and after a failure event occurred. This procedure can be used by both the gNB-DU and the gNB-CU.

The F1 setup function allows to exchange application level data needed for the gNB-DU and gNB-CU to interoperate correctly on the F1 interface. The F1 setup is initiated by the gNB-DU.

The gNB-CU configuration update and gNB-DU configuration update functions allow to update application level configuration data needed between gNB-CU and gNB-DU to interoperate correctly over the F1 interface, and may activate or deactivate cells.

The F1 setup and gNB-DU configuration update functions allow to inform the single network slice selection assistance information (S-NSSAI) supported by the gNB-DU.

The F1 resource coordination function is used to transfer information about frequency resource sharing between gNB-CU and gNB-DU.

(2) System Information Management Function

Scheduling of system broadcast information is carried out in the gNB-DU. The gNB-DU is responsible for transmitting the system information according to the scheduling parameters available.

The gNB-DU is responsible for the encoding of NR master information block (MIB). In case broadcast of system information block type-1 (SIB1) and other SI messages is needed, the gNB-DU is responsible for the encoding of SIB1 and the gNB-CU is responsible for the encoding of other SI messages.

(3) F1 UE Context Management Function

The F1 UE context management function supports the establishment and modification of the necessary overall UE context.

The establishment of the F1 UE context is initiated by the gNB-CU and accepted or rejected by the gNB-DU based on admission control criteria (e.g., resource not available).

The modification of the F1 UE context can be initiated by either gNB-CU or gNB-DU. The receiving node can accept or reject the modification. The F1 UE context management function also supports the release of the context previously established in the gNB-DU. The release of the context is triggered by the gNB-CU either directly or following a request received from the gNB-DU. The gNB-CU request the gNB-DU to release the UE Context when the UE enters RRC_IDLE or RRC_INACTIVE.

This function can be also used to manage DRBs and SRBs, i.e., establishing, modifying and releasing DRB and SRB resources. The establishment and modification of DRB resources are triggered by the gNB-CU and accepted/rejected by the gNB-DU based on resource reservation information and QoS information to be provided to the gNB-DU. For each DRB to be setup or modified, the S-NSSAI may be provided by gNB-CU to the gNB-DU in the UE context setup procedure and the UE context modification procedure.

The mapping between QoS flows and radio bearers is performed by gNB-CU and the granularity of bearer related management over F1 is radio bearer level. For NG-RAN, the gNB-CU provides an aggregated DRB QoS profile and QoS

flow profile to the gNB-DU, and the gNB-DU either accepts the request or rejects it with appropriate cause value. To support packet duplication for intra-gNB-DU carrier aggregation (CA), one data radio bearer should be configured with two GPRS tunneling protocol (GTP)-U tunnels between gNB-CU and a gNB-DU.

With this function, gNB-CU requests the gNB-DU to setup or change of the special cell (SpCell) for the UE, and the gNB-DU either accepts or rejects the request with appropriate cause value.

With this function, the gNB-CU requests the setup of the secondary cell(s) (SCell(s)) at the gNB-DU side, and the gNB-DU accepts all, some or none of the SCell(s) and replies to the gNB-CU. The gNB-CU requests the removal of the SCell(s) for the UE.

(4) RRC Message Transfer Function

This function allows to transfer RRC messages between gNB-CU and gNB-DU. RRC messages are transferred over F1-C. The gNB-CU is responsible for the encoding of the dedicated RRC message with assistance information provided by gNB-DU.

(5) Paging Function

The gNB-DU is responsible for transmitting the paging information according to the scheduling parameters provided.

The gNB-CU provides paging information to enable the gNB-DU to calculate the exact paging occasion (PO) and paging frame (PF). The gNB-CU determines the paging assignment (PA). The gNB-DU consolidates all the paging records for a particular PO, PF and PA, and encodes the final RRC message and broadcasts the paging message on the respective PO, PF in the PA.

(6) Warning Messages Information Transfer Function

This function allows to cooperate with the warning message transmission procedures over NG interface. The gNB-CU is responsible for encoding the warning related SI message and sending it together with other warning related information for the gNB-DU to broadcast over the radio interface.

FIG. 8 shows an interface protocol structure for F1-C to which technical features of the present disclosure can be applied.

A transport network layer (TNL) is based on Internet protocol (IP) transport, comprising a stream control transmission protocol (SCTP) layer on top of the IP layer. An application layer signaling protocol is referred to as an F1 application protocol (F1AP).

Hereinafter, (i) UE Context Modification procedure and (ii) UE Context Modification initiated by gNB-CU are described. Section 8.3 of 3GPP TS 38.473 v16.1.0 may be referred.

The purpose of the UE Context Modification procedure is to modify the established UE Context, e.g., establishing, modifying and releasing radio resources. This procedure is also used to command the gNB-DU to stop data transmission for the UE for mobility. The procedure uses UE-associated signalling.

The UE CONTEXT MODIFICATION REQUEST message is initiated by the gNB-CU.

Upon reception of the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall perform the modifications, and if successful reports the update in the UE CONTEXT MODIFICATION RESPONSE message.

The gNB-DU shall report to the gNB-CU, in the UE CONTEXT MODIFICATION RESPONSE message, the result for all the requested or modified DRBs and SRBs in the following way:

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A list of DRBs which are successfully established shall be included in the DRB Setup List IE;
 A list of DRBs which failed to be established shall be included in the DRB Failed to be Setup List IE;
 A list of DRBs which are successfully modified shall be included in the DRB Modified List IE;
 A list of DRBs which failed to be modified shall be included in the DRB Failed to be Modified List IE;
 A list of SRBs which failed to be established shall be included in the SRB Failed to be Setup List IE.
 A list of successfully established SRBs with logical channel identities for primary path shall be included in the SRB Setup List IE only if CA based PDCP duplication is initiated for the concerned SRBs.
 A list of successfully modified SRBs with logical channel identities for primary path shall be included in the SRB Modified List IE only if CA based PDCP duplication is initiated for the concerned SRBs.

When the gNB-DU reports the unsuccessful establishment of a DRB or SRB, the cause value should be precise enough to enable the gNB-CU to know the reason for the unsuccessful establishment.

In case none of the requested modifications of the UE context can be successfully performed, the gNB-DU shall respond with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

If the gNB-DU is not able to accept the SpCell ID IE in UE CONTEXT MODIFICATION REQUEST message, it shall reply with the UE CONTEXT MODIFICATION FAILURE message.

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a E-UTRAN QoS IE for a GBR QoS DRB but where the GBR QoS Information IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the DRB Failed to Setup List IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a DRB QoS IE for a GBR QoS DRB but where the GBR QoS Flow Information IE is not present, the gNB-DU shall report the establishment of the corresponding DRBs as failed in the DRB Failed to Setup List IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the Delay Critical IE is included in the Dynamic 5QI Descriptor IE within the DRB QoS IE in the UE CONTEXT MODIFICATION REQUEST message and is set to the value "delay critical" but the Maximum Data Burst Volume IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the DRB Failed to Setup List IE of the of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

The purpose of the UE Context Modification procedure is to modify the established UE Context, e.g., establishing, modifying and releasing radio resources. This procedure is also used to command the gNB-DU to stop data transmission for the UE for mobility. The procedure uses UE-associated signalling.

The UE CONTEXT MODIFICATION REQUEST message is initiated by the gNB-CU.

Upon reception of the UE CONTEXT MODIFICATION REQUEST message, the gNB-DU shall perform the modifications, and if successful reports the update in the UE CONTEXT MODIFICATION RESPONSE message.

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The gNB-DU shall report to the gNB-CU, in the UE CONTEXT MODIFICATION RESPONSE message, the result for all the requested or modified DRBs and SRBs in the following way:

A list of DRBs which are successfully established shall be included in the DRB Setup List IE;
 A list of DRBs which failed to be established shall be included in the DRB Failed to be Setup List IE;
 A list of DRBs which are successfully modified shall be included in the DRB Modified List IE;
 A list of DRBs which failed to be modified shall be included in the DRB Failed to be Modified List IE;
 A list of SRBs which failed to be established shall be included in the SRB Failed to be Setup List IE.
 A list of successfully established SRBs with logical channel identities for primary path shall be included in the SRB Setup List IE only if CA based PDCP duplication is initiated for the concerned SRBs.
 A list of successfully modified SRBs with logical channel identities for primary path shall be included in the SRB Modified List IE only if CA based PDCP duplication is initiated for the concerned SRBs.

When the gNB-DU reports the unsuccessful establishment of a DRB or SRB, the cause value should be precise enough to enable the gNB-CU to know the reason for the unsuccessful establishment.

In case none of the requested modifications of the UE context can be successfully performed, the gNB-DU shall respond with the UE CONTEXT MODIFICATION FAILURE message with an appropriate cause value.

If the gNB-DU is not able to accept the SpCell ID IE in UE CONTEXT MODIFICATION REQUEST message, it shall reply with the UE CONTEXT MODIFICATION FAILURE message.

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a E-UTRAN QoS IE for a GBR QoS DRB but where the GBR QoS Information IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the DRB Failed to Setup List IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the gNB-DU receives a UE CONTEXT MODIFICATION REQUEST message containing a DRB QoS IE for a GBR QoS DRB but where the GBR QoS Flow Information IE is not present, the gNB-DU shall report the establishment of the corresponding DRBs as failed in the DRB Failed to Setup List IE of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

If the Delay Critical IE is included in the Dynamic 5QI Descriptor IE within the DRB QoS IE in the UE CONTEXT MODIFICATION REQUEST message and is set to the value "delay critical" but the Maximum Data Burst Volume IE is not present, the gNB-DU shall report the establishment of the corresponding DRB as failed in the DRB Failed to Setup List IE of the of the UE CONTEXT MODIFICATION RESPONSE message with an appropriate cause value.

FIG. 9 shows an example of a method for a Session Leave procedure to which implementations of the present disclosure is applied.

In step S900, there may be potential decision on application level for UE to leave the group.

In step S901, there may be a media stream before the UE has left. UE may receive the media by PTM or PTP.

In step S902, the may UE send an Uplink Non-Access Stratum (UL NAS) Multicast and/or Broadcast (MB) Session Leave Request message (for example, the UL NAS MB Session Leave Request message includes a Temporary

Mobile Group Identity (TMGI)) to the Access and Mobility management Function (AMF). The AMF may remove the TMGI from the UE Context.

In step S903, the AMF may create a Downlink (DL) NAS MB Session Leave Response message and piggy back that on an N2 MB Session Leave message (for example, the N2 MB Session Leave message includes an NGAP ID and/or a TMGI). The NG-RAN may remove the TMGI from the UE context in NG-RAN.

In step S904, the NG-RAN adjusts the PTM and/or PTP transmission if necessary.

In step S905, if this UE is the last UE in this NG-RAN using the MB Session, the NG-RAN may send a Leave message (for example, the Leave message includes LL MC address) to stop the media stream to this NG-RAN node and then delete the MB Session Core Network and Terminals (Ctx).

In step S906, if this UE is the last UE in this AMF being part of the MB Session, the AMF may send an MB Session Release Request (for example, the MB Session Release Request may include TMGI and/or AMF ID) to the MB-SMF, for AMF to unsubscribe to the MB Session. The MB-SMF may remove the AMF in the MB-SMF MB Session Ctx.

In step S907, the MB-SMF may send an MB Session Release Response message to the AMF. AMF may delete its MB Session Ctx.

Meanwhile, a RAN node (for example, a gNB) may be composed of a Central Unit (CU) and a Distributed Unit (DU). For Multicast and/or Broadcast Services, one or more necessary coordination functions (for example, functions hosted by MCE) may be reside in a Central Unit (CU) of a RAN node.

Referring to FIG. 9, a session leave procedure may be used by UEs to inform a network that the UE interest in an MB session has ceased.

In the session leave procedure, when a RAN node may be composed of a CU-Control Plane (CP), a CU-User Plane (UP), and a DU, signalling between the CU-CP and the CU-UP in order to request the MB-User Plane Function (UPF) to stop the media stream to above the RAN node is needed.

Therefore, studies for stopping multicast and broadcast service in a wireless communication system are needed.

Hereinafter, a method for stopping multicast and broadcast service in a wireless communication system, according to some embodiments of the present disclosure, will be described.

FIG. 10 shows an example of a method performed by a Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node in a wireless communication system.

In step S1001, the CU-UP of the RAN node may receive, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure. The first message may include (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication.

For example, the identity for the MB service may include a Temporary Mobile Group Identity (TMGI).

For example, the first message may include a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node. The identity for the MB service may indicate the bearer established for the MB service between the CU-UP and the DU.

For example, the Transmission Stop Indication may include a low layer multicast (LL MC) address for the MB service. The LL MC address for the MB service may be provided by an Access and Mobility management Function (AMF).

According to some embodiments of the present disclosure, the LL MC address may be used for a multicast transport. For example, the LL MC address may be used for a source specific multicast transport. For example, the parameter "LL MC Address" may be accompanied by a "Source host address" parameter.

For example, MB-Session Management Function (SMF) may allocate a TMGI, a Lower Layer Multicast IP Address (LL MC address) for N3, and N6 tunnel information and store the information in a new MB Session Context set to 'inactive' state. MB-SMF may return the TMGI and the N6 tunnel information to the NEF/MBSF. If MB-SMF makes the TMGI allocation, the MB-SMF may allocate a TMGI from a pre-configured TMGI range.

For example, for large networks or for redundancy reasons, the NEF/MBSF may use multiple MB-SMFs (and MB-UPFs).

For example, since the N3 LL MC address is used for 5GS internal transport, it may be considered sufficient to allocate the N3 LL MC address at Session Start, for example, when resources for the MB Session is allocated in different nodes. A one-to-one relationship between N3 LL MC Address and MB Session may be used.

According to some embodiments of the present disclosure, the LL MC address may be used for a Session Start procedure. For example, MB-SMF may set the MB Session Context to active and send MB Session Start messages (for example, the MB Session Start messages may include TMGI, LL MC Address and source host address, 5G Authorized QoS Profile) to all AMFs that has earlier joined the MB Session.

For example, the LL MC Address and source host address of MB-UPF may be provided if multicast transport is configured to be used between the NG-RAN and MB-UPF nodes.

According to some embodiments of the present disclosure, the LL MC address may be used as a transport network IP Multicast Address for MBMS Session Start Procedure for E-UTRAN and UTRAN for EPS. For example, the MME or SGSN may create an MBMS bearer context. The MME/SGSN may store the session attributes and send a Session Start Request message including the session attributes (TMGI, QoS, MBMS service area, list of cell IDs if available, Session identifier, estimated session duration, broadcast (for UTRAN only), transport network IP Multicast Address, IP address of the multicast source, C-TEID, . . .) to E-UTRAN/UTRAN. Optionally in the case of E-UTRAN access and if the MBMS GW supports both IPv4 and IPv6, the MME may include both an IPv4 and an IPv6 IP Multicast address together with the corresponding alternative IP address(es) of the multicast source. When connected to multiple MCEs, the MME should filter the distribution of Session Control message to the MCEs based on the MBMS service area.

In step S1002, the CU-UP of the RAN node may perform the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication.

For example, the CU-UP may release the bearer established for the MB service between the CU-UP and the DU. For example, the CU-UP may release the bearer established

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for the MB service between the CU-UP and the DU, when the release request for the bearer is included in the first message. For example, the CU-CP may release the bearer indicated by the identity for the MB service.

For example, in the MLD and/or IGMP Leave procedure, the CU-UP may transmit, to the MB-UPF, an MLD and/or IGMP Leave message including the LL MC address.

For example, in the MLD and/or IGMP Leave procedure, the CU-UP may leave a multicast group for a MB session indicated by the identity for the MB service.

According to some embodiments of the present disclosure, a CU-UP may receive, from a CU-CP, a multiple Transmission Stop Indications for a same MB service. In this case, the CU-UP may transmit, to a MB-UPF, a single MLD and/or IGMP Leave message for the MB service.

In other words, a CU-UP may receive, from the CU-CP of the RAN node, a third message to initiate the MLD and/or IGMP Leave procedure. The third message may include (i) the identity for the MB service, and/or (ii) a Transmission Stop Indication, before performing the MLD and/or IGMP Leave procedure.

In this case, in the MLD and/or IGMP Leave procedure, the CU-UP may transmit, to the MB-UPF, a single MLD and/or IGMP Leave message corresponding to both the first message and the second message.

According to some embodiments of the present disclosure, there are different versions of IGMP. For example, table 3 shows IGMP versions.

TABLE 3

IGMP Version	Description
IGMPv1	Provides the basic query-response mechanism that allows the multicast device to determine which multicast groups are active and other processes that enable hosts to join and leave a multicast group.
IGMPv2	Extends IGMP, allowing such capabilities as the IGMP leave process, group-specific queries, and an explicit maximum response time field. IGMPv2 also adds the capability for devices to elect the IGMP querier without dependence on the multicast protocol to perform this task.
IGMPv3	Provides for source filtering, which enables a multicast receiver host to signal to a device which groups it wants to receive multicast traffic from, and from which sources this traffic is expected. In addition, IGMPv3 supports the link local address 224.0.0.22, which is the destination IP address for IGMPv3 membership reports; all IGMPv3-capable multicast devices must listen to this address.

For example, a RAN node may use a different IGMP Leave Procedure depending on the version of IGMP.

For example, a RAN node may use IGMPv1 Leave Process.

There is no leave-group message in IGMPv1 to notify the devices on the subnet that a host no longer wants to receive the multicast traffic from a specific group. The host (for example, the RAN node) simply stops processing traffic for the multicast group and ceases responding to IGMP queries with IGMP membership reports for the group. As a result, the only way IGMPv1 devices know that there are no longer any active receivers for a particular multicast group on a subnet is when the devices stop receiving membership reports. To facilitate this process, IGMPv1 devices associate a countdown timer with an IGMP group on a subnet. When a membership report is received for the group on the subnet, the timer is reset. For IGMPv1 devices, this timeout interval is typically three times the query interval (3 minutes). This timeout interval means that the device may continue to

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forward multicast traffic onto the subnet for up to 3 minutes after all hosts have left the multicast group.

For example, a RAN node may use IGMPv2 Leave Process.

IGMPv2 incorporates a leave-group message that provides the means for a host to indicate that it wishes to stop receiving multicast traffic for a specific group. When an IGMPv2 host leaves a multicast group, if it was the last host to respond to a query with a membership report for that group, it sends a leave-group message to the all-devices multicast group (224.0.0.2).

For example, a RAN node may use IGMPv3 Leave Process

IGMPv3 enhances the leave process by introducing the capability for a host to stop receiving traffic from a particular group, source, or channel in IGMP by including or excluding sources, groups, or channels in IGMPv3 membership reports.

In step S1003, the CU-UP of the RAN node may transmit, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

For example, the second message may include the identity for the MB service, which indicates the released bearer for the MB service.

According to some embodiments of the present disclosure, the first message in step S1001 may be a Bearer Context Modification Request message, and the second message in step S1003 may be a Bearer Context Modification Response message.

For example, when a CU-CP receives an MB Session Leave message or an MB Session Resource Release Request message from an AMF, the CU-CP may transmit, to the CU-UP, the Bearer Context Modification Request message, as the first message in step S1001. In this case, the CU-UP may transmit, to the CU-CP, a Bearer Context Modification Response message, as the second message in step S1003.

According to some embodiments of the present disclosure, the first message in step S1001 may be a MB Release Request message, and the second message in step S1003 may be a MB Release Response message.

For example, when a CU-CP receives an MB Session Leave message or an MB Session Resource Release Request message from an AMF, a CU-CP may transmit, to a CU-UP, an MB Release Request message, as the first message in step S1001. In this case, the CU-UP may transmit, to the CU-CP, an MB Release Response message, as the second message in step S1003.

FIGS. 11A and 11B show an example of a method for stopping multicast and broadcast service in a wireless communication system.

In particular, FIGS. 11A and 11B illustrate an example of procedure for session leave, stop, and delete.

In FIGS. 11A and 11B, a CU-CP may provide a CU-UP with the MBS Identity (for example, TMGI) and a stop indication so that the CU-UP requests the MB-UPF to stop the multicast and/or broadcast data transmission for indicated MBS.

In step S1101, after the AMF receives the UL NAS MB Session Leave Request message with the Multicast and/or Broadcast Service (MBS) Identity (for example, TMGI) from the UE, the AMF may create the DL NAS MB Session Leave Response message and piggy back that on the NGAP MB Session Leave message.

In case the AMF receives the MB Session Stop Request message from the MB-Session Management Function

(SMF), the AMF may transmit the MB Session Resource Release Request message with the MBS Identity to the gNB-CU-CP.

If the AMF receives the MB Session Delete Request message from the MB-SMF and the MB session context in the AMF is still in active state, the AMF may send the MB Session Resource Release Request message with the MBS Identity to the gNB-CU-CP.

When the gNB-CU-CP receives the message from the AMF in step S1101, according to whether multicast data transmission between the gNB-DU and the UEs is point-to-point (PTP) or point-to-multipoint (PTM), different procedures may be performed.

According to some embodiments of the present disclosure, in a PTP transmission case, steps S1102a to S1109a may be performed for a procedure for session leave, stop, and delete.

In step S1102a, when the gNB-CU-CP receives the MB Session Leave message and the UE is the last UE in it using the MBS, or the gNB-CU-CP receives the MB Session Resource Release Request message, the gNB-CU-CP may transmit the Bearer Context Modification Request or new message to the gNB-CU-UP to request the release of bearer (for example, data radio bearer or unicast bearer) established for multicast and/or broadcast data transmission between the gNB-CU-UP and the gNB-DU.

This message may include the MBS Identity to indicate that the requested bearer is related to which MBS. This message may contain the Transmission Stop Indication to stop the multicast and/or broadcast data transmission for indicated MBS to the gNB-CU-UP. This Transmission Stop Indication may be the low layer multicast (LL MC) address.

In step S1103a, on receiving the message from the gNB-CU-CP, the gNB-CU-UP may send the MLD/IGMP Leave message with the LL MC address to the MB-UPF and releases the requested bearer. If the gNB-CU-UP receives multiple Bearer Context Modification Request messages for the same MBS, the gNB-CU-UP may only perform sending the MLD/IGMP Leave message once.

In step S1104a, the gNB-CU-UP may transmit the Bearer Context Modification Response or new message to the gNB-CU-CP.

In step S1105a, upon the receipt of the message from the gNB-CU-UP, the gNB-CU-CP may send the UE Context Modification Request or new message to the gNB-DU in order to request the release of bearer (for example, data radio bearer or unicast bearer) established for multicast/broadcast data transmission between the gNB-DU and the UE. This message may contain the MBS Identity to indicate that the requested bearer is related to which MBS.

In step S1106a, when to receive the modification request message, the gNB-DU may release the requested bearer and then respond with the UE Context Modification Response or new message to the gNB-CU-CP.

In step S1107a, on receiving the message from the gNB-DU, the gNB-CU-CP may send the RRCReconfiguration message to the UE in order to provide the UE with the information related to released bearer.

In step S1108a, the UE may transmit the RRCReconfigurationComplete message to the gNB-CU-CP.

For example, if there are multiple UEs provided with the same MBS, steps S1102a to S1108a may be performed for each of that UE before step S1109 is proceeded.

In step S1109a, the gNB-CU-CP may report successful release of resources for the MBS by transmitting the MB Session Resource Release Response message to the AMF.

According to some embodiments of the present disclosure, in a PTM transmission case, steps S1102b to S1107b may be performed for a procedure for session leave, stop, and delete.

In step S1102b, when the gNB-CU-CP receives the MB Session Leave message and the UE is the last UE in it using the MBS, or the gNB-CU-CP receives the MB Session Resource Release Request message, the gNB-CU-CP may transmit the MB Release Request, existing, or new message to the gNB-CU-UP to request the release of bearer established for multicast and/or broadcast data transmission between the gNB-CU-UP and the gNB-DU.

This message may include the MBS Identity to indicate that the requested bearer is related to which MBS. This message may contain the Transmission Stop Indication to stop the multicast/broadcast data transmission for indicated MBS to the gNB-CU-UP. This Transmission Stop Indication may be the LL MC address.

In step S1103b, on receiving the message from the gNB-CU-CP, the gNB-CU-UP may send the MLD/IGMP Leave message with the LL MC address to the MB-UPF and releases the requested bearer.

In step S1104b, the gNB-CU-UP may transmit the MB Release Response, existing, or new message to the gNB-CU-CP. This message may include the MBS Identity to indicate that the released bearer is related to which MBS.

In step S1105b, upon the receipt of the message from the gNB-CU-UP, the gNB-CU-CP may send the MB Context Release Request, existing, or new message to the gNB-DU in order to request the release of MRB established for multicast and/or broadcast data transmission between the gNB-DU and the UE. This message may contain the MBS Identity to indicate that the requested bearer is related to which MBS.

In step S1106b, when to receive the request message, the gNB-DU may release the MRB and respond with the MB Context Release Response, existing, or new message to the gNB-CU-CP. This message may include the MBS Identity to indicate that the released MRB is related to which MBS.

In step S1107b, the gNB-CU-CP may report successful release of resources for the MBS by transmitting the MB Session Resource Release Response message to the AMF.

Hereinafter, an apparatus for stopping multicast and broadcast service in a wireless communication system, according to some embodiments of the present disclosure, will be described.

For example, a Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node may include a processor, and a memory. For example, referring to FIG. 7, a gNB-CU could be an example of the CU of a RAN node, according to some embodiments of the present disclosure. For example, the gNB-CU may include a gNB-CU-CP and a gNB-CU-UP.

According to some embodiments of the present disclosure, the processor may be configured to be coupled operably with the memory.

The processor may be configured to receive, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure. The first message may include (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication. The processor may be configured to perform the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication. The processor may be configured to trans-

mit, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

For example, the first message may include a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node.

For example, the identity for the MB service included in the first message may indicate the bearer established for the MB service between the CU-UP and the DU.

For example, the processor may be configured to release the bearer established for the MB service between the CU-UP and the DU.

For example, the second message may include the identity for the MB service, which indicates the released bearer for the MB service.

For example, the Transmission Stop Indication includes a low layer multicast (LL MC) address for the MB service. The LL MC address for the MB service may be provided by an Access and Mobility management Function (AMF).

For example, in the MLD and/or IGMP Leave procedure, the processor may be configured to transmit, to the MB-UPF, an MLD and/or IGMP Leave message including the LL MC address.

For example, in the MLD and/or IGMP Leave procedure, the processor may be configured to leave a multicast group for a MB session indicated by the identity for the MB service.

For example, the identity for the MB service may include a Temporary Mobile Group Identity (TMGI).

According to some embodiments of the present disclosure, the processor may be configured to receive, from the CU-CP of the RAN node, a third message to initiate the MLD and/or IGMP Leave procedure. The third message may include (i) the identity for the MB service, and/or (ii) a Transmission Stop Indication, before performing the MLD and/or IGMP Leave procedure.

In this case, in the MLD and/or IGMP Leave procedure, the processor may be configured to transmit, to the MB-UPF, a single MLD and/or IGMP Leave message corresponding to both the first message and the second message.

According to some embodiments of the present disclosure, the first message may be a Bearer Context Modification Request message, and the second message may be a Bearer Context Modification Response message.

According to some embodiments of the present disclosure, the first message may be a MB Release Request message, and the second message may be a MB Release Response message.

Hereinafter, a processor for a Central Unit (CU)-User Plane (UP) for stopping multicast and broadcast service in a wireless communication system, according to some embodiments of the present disclosure, will be described.

The processor may be configured to control the CU-UP to receive, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure. The first message may include (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication. The processor may be configured to control the CU-UP to perform the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication. The processor may be configured to control the CU-UP to transmit, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

For example, the first message may include a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node.

For example, the identity for the MB service included in the first message may indicate the bearer established for the MB service between the CU-UP and the DU.

For example, the processor may be configured to control the CU-UP to release the bearer established for the MB service between the CU-UP and the DU.

For example, the second message may include the identity for the MB service, which indicates the released bearer for the MB service.

For example, the Transmission Stop Indication includes a low layer multicast (LL MC) address for the MB service. The LL MC address for the MB service may be provided by an Access and Mobility management Function (AMF).

For example, in the MLD and/or IGMP Leave procedure, the processor may be configured to control the CU-UP to transmit, to the MB-UPF, an MLD and/or IGMP Leave message including the LL MC address.

For example, in the MLD and/or IGMP Leave procedure, the processor may be configured to control the CU-UP to leave a multicast group for a MB session indicated by the identity for the MB service.

For example, the identity for the MB service may include a Temporary Mobile Group Identity (TMGI).

According to some embodiments of the present disclosure, the processor may be configured to control the CU-UP to receive, from the CU-CP of the RAN node, a third message to initiate the MLD and/or IGMP Leave procedure. The third message may include (i) the identity for the MB service, and/or (ii) a Transmission Stop Indication, before performing the MLD and/or IGMP Leave procedure.

In this case, in the MLD and/or IGMP Leave procedure, the processor may be configured to control the CU-UP to transmit, to the MB-UPF, a single MLD and/or IGMP Leave message corresponding to both the first message and the second message.

According to some embodiments of the present disclosure, the first message may be a Bearer Context Modification Request message, and the second message may be a Bearer Context Modification Response message.

According to some embodiments of the present disclosure, the first message may be a MB Release Request message, and the second message may be a MB Release Response message.

Hereinafter, a non-transitory computer-readable medium has stored thereon a plurality of instructions for stopping multicast and broadcast service in a wireless communication system, according to some embodiments of the present disclosure, will be described.

According to some embodiment of the present disclosure, the technical features of the present disclosure could be embodied directly in hardware, in a software executed by a processor, or in a combination of the two. For example, a method performed by a wireless device in a wireless communication may be implemented in hardware, software, firmware, or any combination thereof. For example, a software may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other storage medium.

Some example of storage medium is coupled to the processor such that the processor can read information from the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the

storage medium may reside in an ASIC. For other example, the processor and the storage medium may reside as discrete components.

The computer-readable medium may include a tangible and non-transitory computer-readable storage medium.

For example, non-transitory computer-readable media may include random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, or any other medium that can be used to store instructions or data structures. Non-transitory computer-readable media may also include combinations of the above.

In addition, the method described herein may be realized at least in part by a computer-readable communication medium that carries or communicates code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer.

According to some embodiment of the present disclosure, a non-transitory computer-readable medium has stored thereon a plurality of instructions. The stored a plurality of instructions may be executed by a processor of a Central Unit (CU)-User Plane (UP).

The stored a plurality of instructions may cause the CU-UP to receive, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure. The first message may include (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication. The stored a plurality of instructions may cause the CU-UP to perform the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication. The stored a plurality of instructions may cause the CU-UP to transmit, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

For example, the first message may include a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node.

For example, the identity for the MB service included in the first message may indicate the bearer established for the MB service between the CU-UP and the DU.

For example, the stored a plurality of instructions may cause the CU-UP to release the bearer established for the MB service between the CU-UP and the DU.

For example, the second message may include the identity for the MB service, which indicates the released bearer for the MB service.

For example, the Transmission Stop Indication includes a low layer multicast (LL MC) address for the MB service. The LL MC address for the MB service may be provided by an Access and Mobility management Function (AMF).

For example, in the MLD and/or IGMP Leave procedure, the stored a plurality of instructions may cause the CU-UP to transmit, to the MB-UPF, an MLD and/or IGMP Leave message including the LL MC address.

For example, in the MLD and/or IGMP Leave procedure, the stored a plurality of instructions may cause the CU-UP to leave a multicast group for a MB session indicated by the identity for the MB service.

For example, the identity for the MB service may include a Temporary Mobile Group Identity (TMGI).

According to some embodiments of the present disclosure, the stored a plurality of instructions may cause the CU-UP to receive, from the CU-CP of the RAN node, a third message to initiate the MLD and/or IGMP Leave procedure.

5 The third message may include (i) the identity for the MB service, and/or (ii) a Transmission Stop Indication, before performing the MLD and/or IGMP Leave procedure.

In this case, in the MLD and/or IGMP Leave procedure, the stored a plurality of instructions may cause the CU-UP to transmit, to the MB-UPF, a single MLD and/or IGMP Leave message corresponding to both the first message and the second message.

According to some embodiments of the present disclosure, the first message may be a Bearer Context Modification Request message, and the second message may be a Bearer Context Modification Response message.

According to some embodiments of the present disclosure, the first message may be a MB Release Request message, and the second message may be a MB Release Response message.

The present disclosure may have various advantageous effects.

According to some embodiments of the present disclosure, a Radio Access Network (RAN) node (for example, a base station such as an eNB or a gNB) could efficiently stop to provide a multicast and broadcast service.

For example, a CU-UP can properly request an MB-UPF to stop multicast and/or broadcast data transmission for a MB service when a CU-CP should delete a MB session context due to session leave, stop, or delete.

For example, a RAN node could efficiently use resource for multicast and/or broadcast service.

For example, a RAN node could prevent that the UE receives multicast and/or broadcast data for the MB service in which it is not interested.

According to some embodiments of the present disclosure, a wireless communication system could provide an efficient solution for stopping a multicast and broadcast service.

Advantageous effects which can be obtained through specific embodiments of the present disclosure are not limited to the advantageous effects listed above. For example, there may be a variety of technical effects that a person having ordinary skill in the related art can understand and/or derive from the present disclosure. Accordingly, the specific effects of the present disclosure are not limited to those explicitly described herein, but may include various effects that may be understood or derived from the technical features of the present disclosure.

Claims in the present disclosure can be combined in a various way. For instance, technical features in method claims of the present disclosure can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims can be combined to be implemented or performed in a method. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in an apparatus. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in a method. Other implementations are within the scope of the following claims.

The invention claimed is:

1. A method performed by a Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node in a wireless communication system, the method comprising,
65 receiving, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener

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Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure, wherein the first message includes (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication;

performing the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication; and transmitting, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

2. The method of claim 1, wherein the first message includes a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node.

3. The method of claim 2, wherein the identity for the MB service included in the first message indicates the bearer established for the MB service between the CU-UP and the DU.

4. The method of claim 2, wherein the method further comprises, releasing the bearer established for the MB service between the CU-UP and the DU.

5. The method of claim 4, wherein the second message includes the identity for the MB service, which indicates the released bearer for the MB service.

6. The method of claim 1, wherein the Transmission Stop Indication includes a low layer multicast (LL MC) address for the MB service.

7. The method of claim 6, wherein the LL MC address for the MB service is provided by an Access and Mobility management Function (AMF).

8. The method of claim 6, wherein the MLD and/or IGMP Leave procedure includes transmitting, to the MB-UPF, an MLD and/or IGMP Leave message including the LL MC address.

9. The method of claim 1, wherein the MLD and/or IGMP Leave procedure includes leaving a multicast group for a MB session indicated by the identity for the MB service.

10. The method of claim 1, wherein the identity for the MB service includes a Temporary Mobile Group Identity (TMGI).

11. The method of claim 1, wherein the method further comprises,

receiving, from the CU-CP of the RAN node, a third message to initiate the MLD and/or IGMP Leave procedure, wherein the third message includes (i) the identity for the MB service, and/or (ii) a Transmission Stop Indication, before performing the MLD and/or IGMP Leave procedure.

12. The method of claim 11, wherein the MLD and/or IGMP Leave procedure including transmitting, to the MB-UPF, a single MLD and/or IGMP Leave message corresponding to both the first message and the second message.

13. The method of claim 1, wherein the first message is a Bearer Context Modification Request message, and the second message is a Bearer Context Modification Response message.

14. The method of claim 1, wherein the first message is a MB Release Request message, and the second message is a MB Release Response message.

15. A Central Unit (CU)-User Plane (UP) of a Radio Access Network (RAN) node configured to operate in a wireless communication system, and comprising:

at least one processor; and

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at least one memory operatively coupled to the at least one processor and storing instructions that, based on being executed by the at least one processor, perform operations comprising:

receiving, from a CU-Control Plane (CP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure, wherein the first message includes (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication;

performing the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication; and

transmitting, to the CU-CP of the RAN node, a second message informing that the MLD and/or IGMP Leave procedure is completed.

16. The CU-UP of the RAN node of claim 15, wherein the first message includes a release request for a bearer established for the MB service between the CU-UP and a Distributed Unit (DU) of the RAN node.

17. The CU-UP of the RAN node of claim 16, wherein the identity for the MB service included in the first message indicates the bearer established for the MB service between the CU-UP and the DU.

18. The CU-UP of the RAN node of claim 16, wherein the operations further comprise:

releasing the bearer established for the MB service between the CU-UP and the DU.

19. The CU-UP of the RAN node of claim 18, wherein the second message includes the identity for the MB service, which indicates the released bearer for the MB service.

20. A wireless device configured to operate in a wireless communication system, and comprising:

a transceiver;

at least one processor operatively coupled to the transceiver; and

at least one memory operatively coupled to the at least one processor and storing instructions that, based on being executed by the at least one processor, perform operations comprising:

receiving, from a Central Unit (CU)-Control Plane (CP) of a Radio Access Network (RAN) node, a Radio Resource Control (RRC) reconfiguration message; and transmitting, to the CU-CP, an RRC reconfiguration complete message,

wherein the CU-CP transmits, to a CU-User Plane (UP) of the RAN node, a first message to initiate a Multicast Listener Discovery (MLD) and/or Internet Group Management Protocol (IGMP) Leave procedure, wherein the first message includes (1) an identity for a Multicast and/or Broadcast (MB) service, and (2) a Transmission Stop Indication,

wherein the CU-UP performs the MLD and/or IGMP Leave procedure for the MB service indicated by the identity for the MB service along with an MB-User Plane Function (UPF) based on the Transmission Stop Indication, and

wherein the CU-CP receives, from the CU-UP of the RAN node, a second message informing that the MLD.

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